

- separable equation, 449
- sequence, 70
 - convergent, 91
 - difference equation, 72
 - divergent, 91
 - Fibonacci, 72
 - general term an, 71
 - geometric, 92
 - graph, 71
 - limit of, 90, 91
 - long-term behavior, 90
 - n th term, 70
 - recursive, 72
 - terms, 70
- series, 95
 - finite geometric, 95
 - infinite, 96
- Shannon index, 627
- shear (by matrix multiplication), 545
- shift of a function (horizontal or vertical), 32
- Sierpinski carpet, 100
- sigma notation, 321, 718
 - rules for, 332, 719
- sine function
 - derivative for, 191
 - domain and range, 706
 - graph, 706
 - limits involving, 132, 133
- singular matrix, 529
- sinusoidal curve, 26
- skew-symmetric matrix, 520
- slope
 - of a curve at a point, 159
 - of a nonvertical line, 692
 - of a secant line, 159, 261
 - of a tangent line, 159, 160
- slope field, 441
- slope-intercept form of the equation of a line, 693
- smallpox, epidemiology of, xxxiv
- snake, fitness function for, 581, 595, 614, 618, 623
- soil contamination by a pollutant, 662
- solid of revolution, 410
 - area of, 410
 - volume of, 410
- solution
 - of a difference equation, 95, 216, 422, 425
 - of predator-prey equations, 461
 - to a system of linear equations, 532, 534
 - of a recursive sequence, 73
 - trivial, 533
- solution curve, 441
- solution set for an inequality, 685
- spatial species distribution, 426
- species–area relationship, 266
 - for bats, 25, 63, 64
 - on an island, 458
 - for reptiles, 30
- species conservation, 215, 239
- species discovery curve, 306
- species' niche, 328, 352, 384
- species richness, xxxv
 - of ants, 14
 - of bats in Mexican caves, 25, 63, 266
 - of forest trees, 193
 - of reptiles, 30
- Speedo LZR racer, ii, 603
- sphere
 - equation of, 492
 - functions of, 569
 - geometric formulas for, RP1
 - volume as model for tumor, 408
- spherical colony, difference equation, 77
- spiral (equilibrium), 637, 646
 - stable vs. unstable, 648
- spiral, logarithmic, ii, 468
- spiral cobwebbing, 301
- sporozoite, 79
- spotted owls, conservation biology of, ii, 528
- square matrix, 515
- Squeeze Theorem for limits, 132
- Stability Criterion for Recursive Sequences, 302
 - proof, 716
- stability properties of the equilibrium at the origin, 650
- stable node, 645
- stable spiral, 648
- stage-structured population, 562
- standard basis vectors, 503
- standard deviation, 274
- standard position of an angle, 700
- stationary point, 620
- step function, 10
- sterile insect technique for population growth control, ii, 371, 679
- Sterner and Elser model for homeostasis, 457
- Sternoptyx diaphana*, 545
- Stiles, W. Stanley, 283
- Stiles-Crawford effect, 283
- stretching a function, 32
- subdiagonal of a matrix, 525
- subpopulation, 652, 662
- Substitution Rule
 - for definite integrals, 358
 - for indefinite integrals, 354, 355
- subtraction formulas, 704, RP2
- subtraction of vectors, algebraic, 500
- succession, ecological, 563
- sum
 - of a finite geometric series, 95
 - of an infinite geometric series, 96
- Sum Law for limits, 125
 - proof, 715
- sum of partial fractions, rational function expressed as, 369
- sum of vectors, 496
- Sum Rule for differentiation, 185
- sums of powers of positive integers, 331, 332
- sunflower, phyllotaxy, of, xxxiii
- Superposition Principle, 642
- surge function, 273, 284
- surgery, 24, 75, 121, 185, 457, 482, 663
 - controlling red blood cell loss during, ii, 222
- survival function, 401, 404, 413
- sustainable harvesting, 139, 290, 294, 298, 306
- swimming speed of fish, 155, 168, 296
- swimsuit, effect on drag, 603
- symmetry
 - of a function, 10
 - of an integrable function, 359
- symmetric functions, 359
- symmetric matrix, 520
- synonymous mutation, 70
- system of autonomous nonlinear differential equations, 665, 666, 668
- system of difference equations, 520
- system of differential equations
 - autonomous, 460
 - coupled, 459
- system of first-order autonomous differential equations, 632
- system of linear differential equations, 632
 - first-order autonomous, 632
 - first-order nonautonomous, 632
 - homogeneous, 632
 - long-term behavior of, 649
 - nongeneric, 638
 - nonhomogeneous, 632
 - two-dimensional, 632
- system of linear equations, 532, 631
 - matrix notation for, 532
 - solution to, 532, 534
- systemic lupus erythematosus, 663, 678
- Table of Integrals, 372, RP6–10
- tabular function, 6

- tadpole
 - predation of, 629
 - weight of, 14, 179
- tangent function
 - derivative for, 199
 - graph, 27
- tangent line, 157
 - defined as a derivative, 161
 - defined as a limit, 159
 - slope of, 159, 160, 161
 - vertical, 174
- tangent line approximation, 230
- tangent plane, 597
 - equation of, 597
- tangent plane approximation, 598, 599, 668
- Taylor polynomial, 235
 - n th-degree, 236
 - second-degree, 236
- Tay-Sachs disease, 284
- TEF (thermic effect of food), 314
- telescoping sum, 720
- TEM (transmission electron micrograph), 315
- temperature-humidity index (humidex), 580, 585, 600
- terminal point of a vector, 496
- terms of a sequence, 70
- test values for solving an inequality, 685
- thermic effect of food (TEF), 314
- third derivative, 176
- three-dimensional rectangular coordinate system, 489
- three-dimensional space, 488
- Thompson, D'Arcy, 545
- time-travel experiment with host-parasite
 - genotype frequencies, xlv, 151, 416, 484, 679
- tissue culture, rate of growth of 457
- total fertility rate in the United States, 242
- trace, 569
- trace and determinant condition for
 - stability, 650, 652
- trace of a matrix, 546
 - related to the stability of the origin, 649
- trachea, contraction during coughing, 258
- tragedy of the commons, 298
- transformations of functions, 31, 32
- transmission electron micrograph (TEM), 315
- translation (shift) of a function, 31
- transpose of a matrix, 515
- tree rings, widths of, 13
- triangle, geometric formulas for, RP1
- Triangle Inequality, 714
- Triangle Inequality for vectors, 504, 512
- Triangle Law, 496
- trigonometric identities, 703, RP2
- trigonometric function(s), 26, 701, RP2
 - continuity of, 142, 143
 - derivatives for, 191, 199, 200, RP5
 - graphs of, 27, 705, RP2
 - limits of, 132, 133
 - periodicity of, 27
- trigonometric ratios, 702, RP2
- trigonometry, review of, 699, RP2
- trivial solution, 533
- trout population, prediction of future
 - size, 401
- tumor, 30
 - metastasis of, 478, 662
 - rate of growth of, 211
 - relative change in size, 238
 - resistance to treatment, 639, 662
 - treatment by radioimmunotherapy, 633, 639
- tumor growth
 - aided by matrix-digesting enzymes, 40
 - model for, 361, 452, 456
 - long-term, 662
- two-compartment mixing model, 283, 458, 661, 663, 664, 678
- two-dimensional plane, 488
- two-dimensional system of linear
 - differential equations, 632
- two-intercept form of an equation of a line, 698
- unit vector, 501
- unstable equilibrium, 433, 635
- unstable node, 645
- unstable spiral, 648
- updating function, 73
- upper triangular matrix, 520
- urea concentration rebound after
 - dialysis, 458
- US health care expenditures, 243
- US national debt, 167
- US population growth, model for, 52, 166
- vaccination, 574, 589, 609, 618
 - and antigenic evolution, 495
 - coverage, 66, 479, 663
- vaccine(s)
 - design of, xxxvi, 495
 - epidemiological model of coverage, 479, 663
- vaccine escape, xxxvii, 514
- value of a function, 3
- van der Pol, Balthasar, 477
- van der Pol equation, 477
- van der Waals equation, 595
- vascular system, branching in, 291
- vector(s), 496
 - addition, 496, 500
 - column, 515
 - complex components of, 557
 - components of, 498, 499
 - difference of, 498
 - displacement, 496
 - equivalent, 496
 - geometric representation, 499
 - initial point of, 496
 - length, 500
 - magnitude, 500
 - n -dimensional, 502
 - normal, 508
 - orthogonal, 506
 - parallel, 497
 - perpendicular, 506
 - position, 499
 - projections of, 508
 - properties of, 502
 - multiplication, 497, 500
 - row, 515
 - scalar multiplication, 497, 500
 - standard basis, 503
 - subtraction, 500
 - sum of, 496
 - terminal point of, 496
 - unit, 501
 - zero, 496
- vector projection, 508, 509
- vectorcardiography, xxxv, 251, 256, 504, 512
 - and Einthoven's triangle, 509
 - matrix model for, 527, 535, 536, 545, 547, 558
- velocity
 - average, 111
 - derivative defined as, 164
 - instantaneous, 112
 - as a limit of a function, 111
 - as a rate of change, 163
- velocity function, 310
- velocity gradient, 187
- ventilation, lung volume during, 457
- Verhulst, Pierre-François, 424
- vertex of a parabola, 694
- vertical asymptote, 120
- Vertical Line Test, 8
- vertical tangent line, 174
- vertical trajectory of a zebra finch, ii, 22
- viral identification, 561
- viral infection, 89
 - model for, 101

802 INDEX

- viral load (function) of HIV, 47, 62, 107, 108, 166
 - exponential model for, 47, 208
- virulence, disease, 110, 258
- virus, 89, 547, 561
- virus clusters in antigenic space, 561
- Volterra, Vito, 461
- volume of a solid, 405, 407
- von Bertalanffy, Ludwig, 451
- von Bertalanffy model for fish length, 110, 215, 295, 351, 400, 419, 429, 450
- walking
 - ground reaction force, 14, 41, 179
 - metabolic power, 10
- washer (annular ring), 409
- wave equation, 596
- Weibull equation for drug dissolution, 429, 456
- whale, right, 521
- wheat, rate of production of, 609
- whelks, dropped by crows, 296
- wind-chill index, 566, 567, 580, 581, 593
 - linear approximation of, 601, 602
 - model of function for, 595
- wingspan, related to weight, 30
- wolf population, model for, ii, 609
- world population growth, model for, 2, 6, 46, 52, 146, 213, 217, 220, 283, 398, 424
- world-record hammer throw distances, 99
- world-record sprint times, 99
- x -coordinate, 689
- y -coordinate, 689
- yeast population, growth of, ii, 179, 244, 420, 430, 448
- yellow perch, matrix diagram for population of, 527
- zebra finch, vertical trajectory of, ii, 22
- zero vector, 496
- zero-order kinetic equation, 429

ALGEBRA

Arithmetic Operations

$$a(b + c) = ab + ac$$

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$$

$$\frac{a + c}{b} = \frac{a}{b} + \frac{c}{b}$$

$$\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

Exponents and Radicals

$$x^m x^n = x^{m+n}$$

$$\frac{x^m}{x^n} = x^{m-n}$$

$$(x^m)^n = x^{mn}$$

$$x^{-n} = \frac{1}{x^n}$$

$$(xy)^n = x^n y^n$$

$$\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n}$$

$$x^{1/n} = \sqrt[n]{x}$$

$$x^{m/n} = \sqrt[n]{x^m} = (\sqrt[n]{x})^m$$

$$\sqrt[n]{xy} = \sqrt[n]{x} \sqrt[n]{y}$$

$$\sqrt[n]{\frac{x}{y}} = \frac{\sqrt[n]{x}}{\sqrt[n]{y}}$$

Factoring Special Polynomials

$$x^2 - y^2 = (x + y)(x - y)$$

$$x^3 + y^3 = (x + y)(x^2 - xy + y^2)$$

$$x^3 - y^3 = (x - y)(x^2 + xy + y^2)$$

Binomial Theorem

$$(x + y)^2 = x^2 + 2xy + y^2$$

$$(x - y)^2 = x^2 - 2xy + y^2$$

$$(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$$

$$(x - y)^3 = x^3 - 3x^2y + 3xy^2 - y^3$$

$$(x + y)^n = x^n + nx^{n-1}y + \frac{n(n-1)}{2}x^{n-2}y^2$$

$$+ \cdots + \binom{n}{k}x^{n-k}y^k + \cdots + nxy^{n-1} + y^n$$

$$\text{where } \binom{n}{k} = \frac{n(n-1) \cdots (n-k+1)}{1 \cdot 2 \cdot 3 \cdots k}$$

Quadratic Formula

$$\text{If } ax^2 + bx + c = 0, \text{ then } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Inequalities and Absolute Value

$$\text{If } a < b \text{ and } b < c, \text{ then } a < c.$$

$$\text{If } a < b, \text{ then } a + c < b + c.$$

$$\text{If } a < b \text{ and } c > 0, \text{ then } ca < cb.$$

$$\text{If } a < b \text{ and } c < 0, \text{ then } ca > cb.$$

$$\text{If } a > 0, \text{ then}$$

$$|x| = a \text{ means } x = a \text{ or } x = -a$$

$$|x| < a \text{ means } -a < x < a$$

$$|x| > a \text{ means } x > a \text{ or } x < -a$$

GEOMETRY

Geometric Formulas

Formulas for area A , circumference C , and volume V :

Triangle

$$A = \frac{1}{2}bh$$

$$= \frac{1}{2}ab \sin \theta$$

Circle

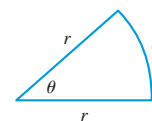
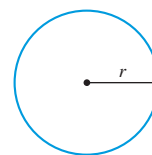
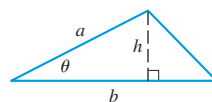
$$A = \pi r^2$$

$$C = 2\pi r$$

Sector of Circle

$$A = \frac{1}{2}r^2\theta$$

$$s = r\theta \text{ (}\theta \text{ in radians)}$$



Sphere

$$V = \frac{4}{3}\pi r^3$$

$$A = 4\pi r^2$$

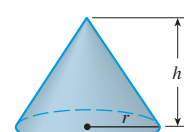
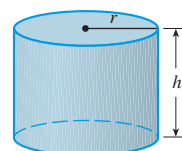
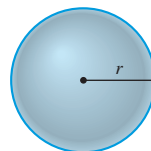
Cylinder

$$V = \pi r^2 h$$

Cone

$$V = \frac{1}{3}\pi r^2 h$$

$$A = \pi r \sqrt{r^2 + h^2}$$



Distance and Midpoint Formulas

Distance between $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\text{Midpoint of } \overline{P_1P_2}: \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Lines

Slope of line through $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-slope equation of line through $P_1(x_1, y_1)$ with slope m :

$$y - y_1 = m(x - x_1)$$

Slope-intercept equation of line with slope m and y-intercept b :

$$y = mx + b$$

Circles

Equation of the circle with center (h, k) and radius r :

$$(x - h)^2 + (y - k)^2 = r^2$$

TRIGONOMETRY

Angle Measurement

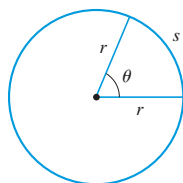
$$\pi \text{ radians} = 180^\circ$$

$$1^\circ = \frac{\pi}{180} \text{ rad}$$

$$1 \text{ rad} = \frac{180^\circ}{\pi}$$

$$s = r\theta$$

(θ in radians)



Right Angle Trigonometry

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

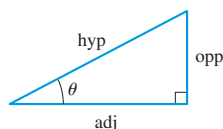
$$\csc \theta = \frac{\text{hyp}}{\text{opp}}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

$$\sec \theta = \frac{\text{hyp}}{\text{adj}}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$

$$\cot \theta = \frac{\text{adj}}{\text{opp}}$$



Trigonometric Functions

$$\sin \theta = \frac{y}{r}$$

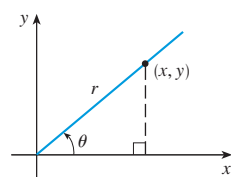
$$\csc \theta = \frac{r}{y}$$

$$\cos \theta = \frac{x}{r}$$

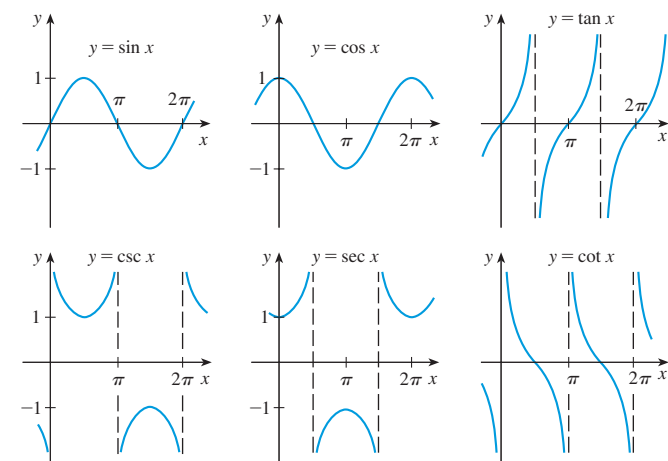
$$\sec \theta = \frac{r}{x}$$

$$\tan \theta = \frac{y}{x}$$

$$\cot \theta = \frac{x}{y}$$



Graphs of Trigonometric Functions



Trigonometric Functions of Important Angles

θ	radians	$\sin \theta$	$\cos \theta$	$\tan \theta$
0°	0	0	1	0
30°	$\pi/6$	$1/2$	$\sqrt{3}/2$	$\sqrt{3}/3$
45°	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1
60°	$\pi/3$	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$
90°	$\pi/2$	1	0	—

Fundamental Identities

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(-\theta) = -\sin \theta$$

$$\cos(-\theta) = \cos \theta$$

$$\tan(-\theta) = -\tan \theta$$

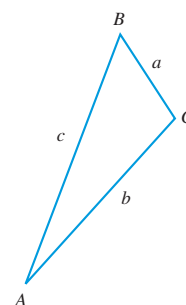
$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta$$

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$$

$$\tan\left(\frac{\pi}{2} - \theta\right) = \cot \theta$$

The Law of Sines

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$



The Law of Cosines

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Addition and Subtraction Formulas

$$\sin(x + y) = \sin x \cos y + \cos x \sin y$$

$$\sin(x - y) = \sin x \cos y - \cos x \sin y$$

$$\cos(x + y) = \cos x \cos y - \sin x \sin y$$

$$\cos(x - y) = \cos x \cos y + \sin x \sin y$$

$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

$$\tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

Double-Angle Formulas

$$\sin 2x = 2 \sin x \cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

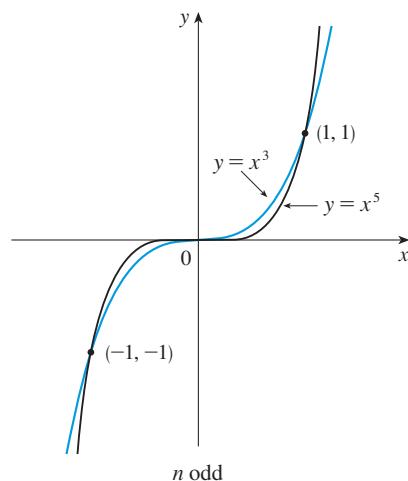
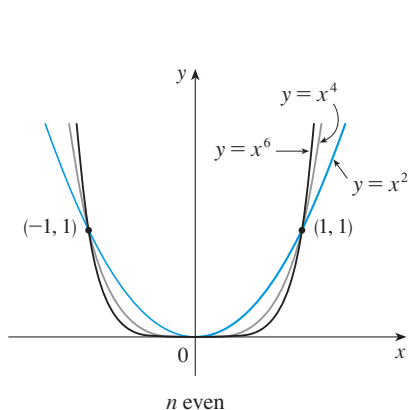
Half-Angle Formulas

$$\sin^2 x = \frac{1 - \cos 2x}{2} \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

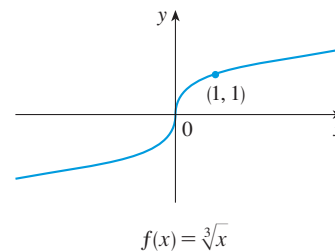
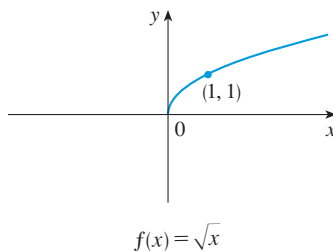
SPECIAL FUNCTIONS

Power Functions $f(x) = x^p$

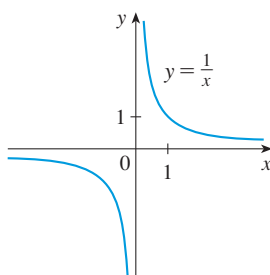
(i) $f(x) = x^n$, n a positive integer



(ii) $f(x) = x^{1/n} = \sqrt[n]{x}$, n a positive integer



(iii) $f(x) = x^{-1} = \frac{1}{x}$

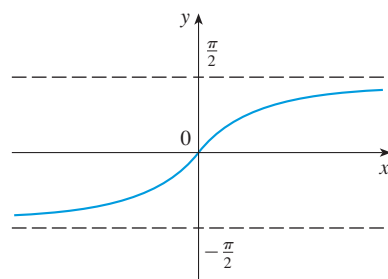


Inverse Trigonometric Functions

$$\arcsin x = \sin^{-1}x = y \iff \sin y = x \quad \text{and} \quad -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

$$\arccos x = \cos^{-1}x = y \iff \cos y = x \quad \text{and} \quad 0 \leq y \leq \pi$$

$$\arctan x = \tan^{-1}x = y \iff \tan y = x \quad \text{and} \quad -\frac{\pi}{2} < y < \frac{\pi}{2}$$



$$y = \tan^{-1}x = \arctan x$$

$$\lim_{x \rightarrow -\infty} \tan^{-1}x = -\frac{\pi}{2}$$

$$\lim_{x \rightarrow \infty} \tan^{-1}x = \frac{\pi}{2}$$

SPECIAL FUNCTIONS

Exponential and Logarithmic Functions

$$\log_b x = y \iff b^y = x$$

$$\ln x = \log_e x, \text{ where } \ln e = 1$$

$$\ln x = y \iff e^y = x$$

Cancellation Equations

$$\log_b(b^x) = x \quad b^{\log_b x} = x$$

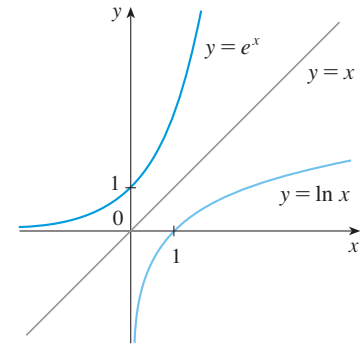
$$\ln(e^x) = x \quad e^{\ln x} = x$$

Laws of Logarithms

$$1. \log_b(xy) = \log_b x + \log_b y$$

$$2. \log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y$$

$$3. \log_b(x^r) = r \log_b x$$

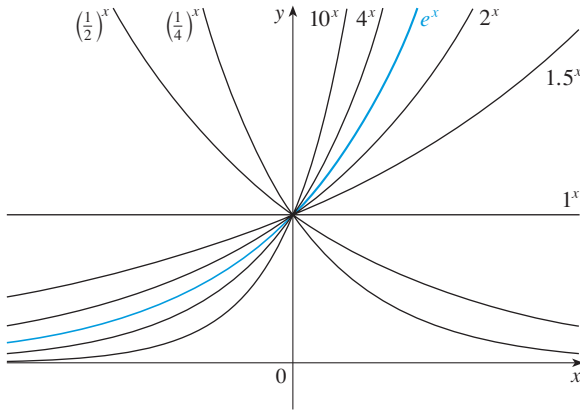


$$\lim_{x \rightarrow -\infty} e^x = 0$$

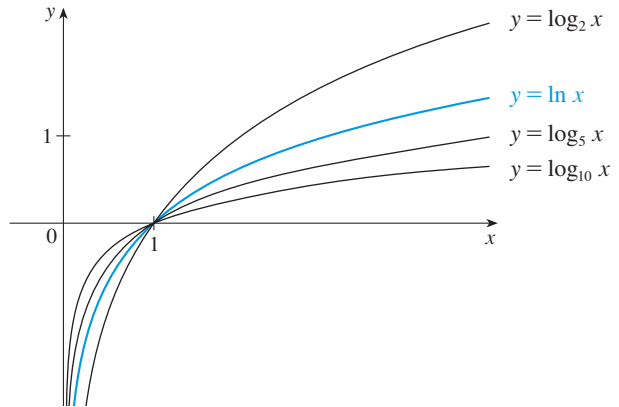
$$\lim_{x \rightarrow \infty} e^x = \infty$$

$$\lim_{x \rightarrow 0^+} \ln x = -\infty$$

$$\lim_{x \rightarrow \infty} \ln x = \infty$$



Exponential functions



Logarithmic functions

Hyperbolic Functions

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

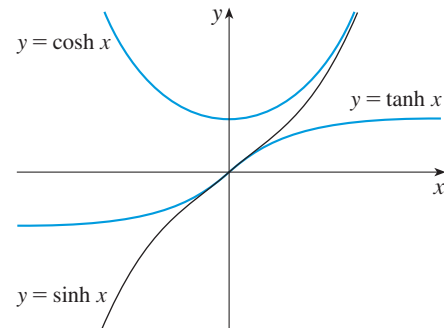
$$\operatorname{csch} x = \frac{1}{\sinh x}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\operatorname{sech} x = \frac{1}{\cosh x}$$

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\coth x = \frac{\cosh x}{\sinh x}$$



Inverse Hyperbolic Functions

$$y = \sinh^{-1} x \iff \sinh y = x$$

$$\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$$

$$y = \cosh^{-1} x \iff \cosh y = x \text{ and } y \geq 0$$

$$\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1})$$

$$y = \tanh^{-1} x \iff \tanh y = x$$

$$\tanh^{-1} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)$$

DIFFERENTIATION RULES

General Formulas

1. $\frac{d}{dx}(c) = 0$
2. $\frac{d}{dx}[cf(x)] = cf'(x)$
3. $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$
4. $\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x)$
5. $\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$ (Product Rule)
6. $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$ (Quotient Rule)
7. $\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$ (Chain Rule)
8. $\frac{d}{dx}(x^n) = nx^{n-1}$ (Power Rule)

Exponential and Logarithmic Functions

9. $\frac{d}{dx}(e^x) = e^x$
10. $\frac{d}{dx}(b^x) = b^x \ln b$
11. $\frac{d}{dx} \ln |x| = \frac{1}{x}$
12. $\frac{d}{dx}(\log_b x) = \frac{1}{x \ln b}$

Trigonometric Functions

13. $\frac{d}{dx}(\sin x) = \cos x$
14. $\frac{d}{dx}(\cos x) = -\sin x$
15. $\frac{d}{dx}(\tan x) = \sec^2 x$
16. $\frac{d}{dx}(\csc x) = -\csc x \cot x$
17. $\frac{d}{dx}(\sec x) = \sec x \tan x$
18. $\frac{d}{dx}(\cot x) = -\csc^2 x$

Inverse Trigonometric Functions

19. $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$
20. $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$
21. $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$
22. $\frac{d}{dx}(\csc^{-1} x) = -\frac{1}{x\sqrt{x^2-1}}$
23. $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$
24. $\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$

Hyperbolic Functions

25. $\frac{d}{dx}(\sinh x) = \cosh x$
26. $\frac{d}{dx}(\cosh x) = \sinh x$
27. $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$
28. $\frac{d}{dx}(\operatorname{csch} x) = -\operatorname{csch} x \coth x$
29. $\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$
30. $\frac{d}{dx}(\coth x) = -\operatorname{csch}^2 x$

Inverse Hyperbolic Functions

31. $\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$
32. $\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}$
33. $\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}$
34. $\frac{d}{dx}(\operatorname{csch}^{-1} x) = -\frac{1}{|x|\sqrt{x^2+1}}$
35. $\frac{d}{dx}(\operatorname{sech}^{-1} x) = -\frac{1}{x\sqrt{1-x^2}}$
36. $\frac{d}{dx}(\coth^{-1} x) = \frac{1}{1-x^2}$

TABLE OF INTEGRALS

Basic Form

1. $\int u \, dv = uv - \int v \, du$
2. $\int u^n \, du = \frac{u^{n+1}}{n+1} + C, \quad n \neq -1$
3. $\int \frac{du}{u} = \ln |u| + C$
4. $\int e^u \, du = e^u + C$
5. $\int b^u \, du = \frac{b^u}{\ln b} + C$
6. $\int \sin u \, du = -\cos u + C$
7. $\int \cos u \, du = \sin u + C$
8. $\int \sec^2 u \, du = \tan u + C$
9. $\int \csc^2 u \, du = -\cot u + C$
10. $\int \sec u \tan u \, du = \sec u + C$
11. $\int \csc u \cot u \, du = -\csc u + C$
12. $\int \tan u \, du = \ln |\sec u| + C$
13. $\int \cot u \, du = \ln |\sin u| + C$
14. $\int \sec u \, du = \ln |\sec u + \tan u| + C$
15. $\int \csc u \, du = \ln |\csc u - \cot u| + C$
16. $\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1} \frac{u}{a} + C, \quad a > 0$
17. $\int \frac{du}{a^2 + u^2} = \frac{1}{a} \tan^{-1} \frac{u}{a} + C$
18. $\int \frac{du}{u\sqrt{u^2 - a^2}} = \frac{1}{a} \sec^{-1} \frac{u}{a} + C$
19. $\int \frac{du}{a^2 - u^2} = \frac{1}{2a} \ln \left| \frac{u+a}{u-a} \right| + C$
20. $\int \frac{du}{u^2 - a^2} = \frac{1}{2a} \ln \left| \frac{u-a}{u+a} \right| + C$

Forms Involving $\sqrt{a^2 + u^2}$, $a > 0$

21. $\int \sqrt{a^2 + u^2} \, du = \frac{u}{2} \sqrt{a^2 + u^2} + \frac{a^2}{2} \ln(u + \sqrt{a^2 + u^2}) + C$
22. $\int u^2 \sqrt{a^2 + u^2} \, du = \frac{u}{8} (a^2 + 2u^2) \sqrt{a^2 + u^2} - \frac{a^4}{8} \ln(u + \sqrt{a^2 + u^2}) + C$
23. $\int \frac{\sqrt{a^2 + u^2}}{u} \, du = \sqrt{a^2 + u^2} - a \ln \left| \frac{a + \sqrt{a^2 + u^2}}{u} \right| + C$
24. $\int \frac{\sqrt{a^2 + u^2}}{u^2} \, du = -\frac{\sqrt{a^2 + u^2}}{u} + \ln(u + \sqrt{a^2 + u^2}) + C$
25. $\int \frac{du}{\sqrt{a^2 + u^2}} = \ln(u + \sqrt{a^2 + u^2}) + C$
26. $\int \frac{u^2 \, du}{\sqrt{a^2 + u^2}} = \frac{u}{2} \sqrt{a^2 + u^2} - \frac{a^2}{2} \ln(u + \sqrt{a^2 + u^2}) + C$
27. $\int \frac{du}{u\sqrt{a^2 + u^2}} = -\frac{1}{a} \ln \left| \frac{\sqrt{a^2 + u^2} + a}{u} \right| + C$
28. $\int \frac{du}{u^2 \sqrt{a^2 + u^2}} = -\frac{\sqrt{a^2 + u^2}}{a^2 u} + C$
29. $\int \frac{du}{(a^2 + u^2)^{3/2}} = \frac{u}{a^2 \sqrt{a^2 + u^2}} + C$

TABLE OF INTEGRALS

Forms Involving $\sqrt{a^2 - u^2}$, $a > 0$

30. $\int \sqrt{a^2 - u^2} du = \frac{u}{2} \sqrt{a^2 - u^2} + \frac{a^2}{2} \sin^{-1} \frac{u}{a} + C$
31. $\int u^2 \sqrt{a^2 - u^2} du = \frac{u}{8} (2u^2 - a^2) \sqrt{a^2 - u^2} + \frac{a^4}{8} \sin^{-1} \frac{u}{a} + C$
32. $\int \frac{\sqrt{a^2 - u^2}}{u} du = \sqrt{a^2 - u^2} - a \ln \left| \frac{a + \sqrt{a^2 - u^2}}{u} \right| + C$
33. $\int \frac{\sqrt{a^2 - u^2}}{u^2} du = -\frac{1}{u} \sqrt{a^2 - u^2} - \sin^{-1} \frac{u}{a} + C$
34. $\int \frac{u^2 du}{\sqrt{a^2 - u^2}} = -\frac{u}{2} \sqrt{a^2 - u^2} + \frac{a^2}{2} \sin^{-1} \frac{u}{a} + C$
35. $\int \frac{du}{u \sqrt{a^2 - u^2}} = -\frac{1}{a} \ln \left| \frac{a + \sqrt{a^2 - u^2}}{u} \right| + C$
36. $\int \frac{du}{u^2 \sqrt{a^2 - u^2}} = -\frac{1}{a^2 u} \sqrt{a^2 - u^2} + C$
37. $\int (a^2 - u^2)^{3/2} du = -\frac{u}{8} (2u^2 - 5a^2) \sqrt{a^2 - u^2} + \frac{3a^4}{8} \sin^{-1} \frac{u}{a} + C$
38. $\int \frac{du}{(a^2 - u^2)^{3/2}} = \frac{u}{a^2 \sqrt{a^2 - u^2}} + C$

Forms Involving $\sqrt{u^2 - a^2}$, $a > 0$

39. $\int \sqrt{u^2 - a^2} du = \frac{u}{2} \sqrt{u^2 - a^2} - \frac{a^2}{2} \ln |u + \sqrt{u^2 - a^2}| + C$
40. $\int u^2 \sqrt{u^2 - a^2} du = \frac{u}{8} (2u^2 - a^2) \sqrt{u^2 - a^2} - \frac{a^4}{8} \ln |u + \sqrt{u^2 - a^2}| + C$
41. $\int \frac{\sqrt{u^2 - a^2}}{u} du = \sqrt{u^2 - a^2} - a \cos^{-1} \frac{a}{|u|} + C$
42. $\int \frac{\sqrt{u^2 - a^2}}{u^2} du = -\frac{\sqrt{u^2 - a^2}}{u} + \ln |u + \sqrt{u^2 - a^2}| + C$
43. $\int \frac{du}{\sqrt{u^2 - a^2}} = \ln |u + \sqrt{u^2 - a^2}| + C$
44. $\int \frac{u^2 du}{\sqrt{u^2 - a^2}} = \frac{u}{2} \sqrt{u^2 - a^2} + \frac{a^2}{2} \ln |u + \sqrt{u^2 - a^2}| + C$
45. $\int \frac{du}{u^2 \sqrt{u^2 - a^2}} = \frac{\sqrt{u^2 - a^2}}{a^2 u} + C$
46. $\int \frac{du}{(u^2 - a^2)^{3/2}} = -\frac{u}{a^2 \sqrt{u^2 - a^2}} + C$

TABLE OF INTEGRALS

Forms Involving $a + bu$

$$47. \int \frac{u \, du}{a + bu} = \frac{1}{b^2} (a + bu - a \ln |a + bu|) + C$$

$$48. \int \frac{u^2 \, du}{a + bu} = \frac{1}{2b^3} [(a + bu)^2 - 4a(a + bu) + 2a^2 \ln |a + bu|] + C$$

$$49. \int \frac{du}{u(a + bu)} = \frac{1}{a} \ln \left| \frac{u}{a + bu} \right| + C$$

$$50. \int \frac{du}{u^2(a + bu)} = -\frac{1}{au} + \frac{b}{a^2} \ln \left| \frac{a + bu}{u} \right| + C$$

$$51. \int \frac{u \, du}{(a + bu)^2} = \frac{a}{b^2(a + bu)} + \frac{1}{b^2} \ln |a + bu| + C$$

$$52. \int \frac{du}{u(a + bu)^2} = \frac{1}{a(a + bu)} - \frac{1}{a^2} \ln \left| \frac{a + bu}{u} \right| + C$$

$$53. \int \frac{u^2 \, du}{(a + bu)^2} = \frac{1}{b^3} \left(a + bu - \frac{a^2}{a + bu} - 2a \ln |a + bu| \right) + C$$

$$54. \int u \sqrt{a + bu} \, du = \frac{2}{15b^2} (3bu - 2a)(a + bu)^{3/2} + C$$

$$55. \int \frac{u \, du}{\sqrt{a + bu}} = \frac{2}{3b^2} (bu - 2a) \sqrt{a + bu} + C$$

$$56. \int \frac{u^2 \, du}{\sqrt{a + bu}} = \frac{2}{15b^3} (8a^2 + 3b^2u^2 - 4abu) \sqrt{a + bu} + C$$

$$57. \int \frac{du}{u \sqrt{a + bu}} = \frac{1}{\sqrt{a}} \ln \left| \frac{\sqrt{a + bu} - \sqrt{a}}{\sqrt{a + bu} + \sqrt{a}} \right| + C, \quad \text{if } a > 0$$

$$= \frac{2}{\sqrt{-a}} \tan^{-1} \sqrt{\frac{a + bu}{-a}} + C, \quad \text{if } a < 0$$

$$58. \int \frac{\sqrt{a + bu}}{u} \, du = 2\sqrt{a + bu} + a \int \frac{du}{u \sqrt{a + bu}}$$

$$59. \int \frac{\sqrt{a + bu}}{u^2} \, du = -\frac{\sqrt{a + bu}}{u} + \frac{b}{2} \int \frac{du}{u \sqrt{a + bu}}$$

$$60. \int u^n \sqrt{a + bu} \, du = \frac{2}{b(2n + 3)} \left[u^n (a + bu)^{3/2} - na \int u^{n-1} \sqrt{a + bu} \, du \right]$$

$$61. \int \frac{u^n \, du}{\sqrt{a + bu}} = \frac{2u^n \sqrt{a + bu}}{b(2n + 1)} - \frac{2na}{b(2n + 1)} \int \frac{u^{n-1} \, du}{\sqrt{a + bu}}$$

$$62. \int \frac{du}{u^n \sqrt{a + bu}} = -\frac{\sqrt{a + bu}}{a(n - 1)u^{n-1}} - \frac{b(2n - 3)}{2a(n - 1)} \int \frac{du}{u^{n-1} \sqrt{a + bu}}$$

TABLE OF INTEGRALS

Trigonometric Forms

$$63. \int \sin^2 u \, du = \frac{1}{2}u - \frac{1}{4}\sin 2u + C$$

$$64. \int \cos^2 u \, du = \frac{1}{2}u + \frac{1}{4}\sin 2u + C$$

$$65. \int \tan^2 u \, du = \tan u - u + C$$

$$66. \int \cot^2 u \, du = -\cot u - u + C$$

$$67. \int \sin^3 u \, du = -\frac{1}{3}(2 + \sin^2 u) \cos u + C$$

$$68. \int \cos^3 u \, du = \frac{1}{3}(2 + \cos^2 u) \sin u + C$$

$$69. \int \tan^3 u \, du = \frac{1}{2}\tan^2 u + \ln |\cos u| + C$$

$$70. \int \cot^3 u \, du = -\frac{1}{2}\cot^2 u - \ln |\sin u| + C$$

$$71. \int \sec^3 u \, du = \frac{1}{2}\sec u \tan u + \frac{1}{2}\ln |\sec u + \tan u| + C$$

$$72. \int \csc^3 u \, du = -\frac{1}{2}\csc u \cot u + \frac{1}{2}\ln |\csc u - \cot u| + C$$

$$73. \int \sin^n u \, du = -\frac{1}{n}\sin^{n-1}u \cos u + \frac{n-1}{n}\int \sin^{n-2}u \, du$$

$$74. \int \cos^n u \, du = \frac{1}{n}\cos^{n-1}u \sin u + \frac{n-1}{n}\int \cos^{n-2}u \, du$$

$$75. \int \tan^n u \, du = \frac{1}{n-1}\tan^{n-1}u - \int \tan^{n-2}u \, du$$

$$76. \int \cot^n u \, du = \frac{-1}{n-1}\cot^{n-1}u - \int \cot^{n-2}u \, du$$

$$77. \int \sec^n u \, du = \frac{1}{n-1}\tan u \sec^{n-2}u + \frac{n-2}{n-1}\int \sec^{n-2}u \, du$$

$$78. \int \csc^n u \, du = \frac{-1}{n-1}\cot u \csc^{n-2}u + \frac{n-2}{n-1}\int \csc^{n-2}u \, du$$

$$79. \int \sin au \sin bu \, du = \frac{\sin(a-b)u}{2(a-b)} - \frac{\sin(a+b)u}{2(a+b)} + C$$

$$80. \int \cos au \cos bu \, du = \frac{\sin(a-b)u}{2(a-b)} + \frac{\sin(a+b)u}{2(a+b)} + C$$

$$81. \int \sin au \cos bu \, du = -\frac{\cos(a-b)u}{2(a-b)} - \frac{\cos(a+b)u}{2(a+b)} + C$$

$$82. \int u \sin u \, du = \sin u - u \cos u + C$$

$$83. \int u \cos u \, du = \cos u + u \sin u + C$$

$$84. \int u^n \sin u \, du = -u^n \cos u + n \int u^{n-1} \cos u \, du$$

$$85. \int u^n \cos u \, du = u^n \sin u - n \int u^{n-1} \sin u \, du$$

$$86. \int \sin^n u \cos^m u \, du = -\frac{\sin^{n-1}u \cos^{m+1}u}{n+m} + \frac{n-1}{n+m}\int \sin^{n-2}u \cos^m u \, du$$

$$= \frac{\sin^{n+1}u \cos^{m-1}u}{n+m} + \frac{m-1}{n+m}\int \sin^n u \cos^{m-2}u \, du$$

Inverse Trigonometric Forms

$$87. \int \sin^{-1} u \, du = u \sin^{-1} u + \sqrt{1-u^2} + C$$

$$88. \int \cos^{-1} u \, du = u \cos^{-1} u - \sqrt{1-u^2} + C$$

$$89. \int \tan^{-1} u \, du = u \tan^{-1} u - \frac{1}{2}\ln(1+u^2) + C$$

$$90. \int u \sin^{-1} u \, du = \frac{2u^2-1}{4}\sin^{-1} u + \frac{u\sqrt{1-u^2}}{4} + C$$

$$91. \int u \cos^{-1} u \, du = \frac{2u^2-1}{4}\cos^{-1} u - \frac{u\sqrt{1-u^2}}{4} + C$$

$$92. \int u \tan^{-1} u \, du = \frac{u^2+1}{2}\tan^{-1} u - \frac{u}{2} + C$$

$$93. \int u^n \sin^{-1} u \, du = \frac{1}{n+1}\left[u^{n+1}\sin^{-1} u - \int \frac{u^{n+1} du}{\sqrt{1-u^2}}\right], \quad n \neq -1$$

$$94. \int u^n \cos^{-1} u \, du = \frac{1}{n+1}\left[u^{n+1}\cos^{-1} u + \int \frac{u^{n+1} du}{\sqrt{1-u^2}}\right], \quad n \neq -1$$

$$95. \int u^n \tan^{-1} u \, du = \frac{1}{n+1}\left[u^{n+1}\tan^{-1} u - \int \frac{u^{n+1} du}{1+u^2}\right], \quad n \neq -1$$

TABLE OF INTEGRALS

Exponential and Logarithmic Forms

$$96. \int u e^{au} du = \frac{1}{a^2} (au - 1) e^{au} + C$$

$$97. \int u^n e^{au} du = \frac{1}{a} u^n e^{au} - \frac{n}{a} \int u^{n-1} e^{au} du$$

$$98. \int e^{au} \sin bu du = \frac{e^{au}}{a^2 + b^2} (a \sin bu - b \cos bu) + C$$

$$99. \int e^{au} \cos bu du = \frac{e^{au}}{a^2 + b^2} (a \cos bu + b \sin bu) + C$$

$$100. \int \ln u du = u \ln u - u + C$$

$$101. \int u^n \ln u du = \frac{u^{n+1}}{(n+1)^2} [(n+1) \ln u - 1] + C$$

$$102. \int \frac{1}{u \ln u} du = \ln |\ln u| + C$$

Hyperbolic Forms

$$103. \int \sinh u du = \cosh u + C$$

$$104. \int \cosh u du = \sinh u + C$$

$$105. \int \tanh u du = \ln \cosh u + C$$

$$106. \int \coth u du = \ln |\sinh u| + C$$

$$107. \int \operatorname{sech} u du = \tan^{-1} |\sinh u| + C$$

$$108. \int \operatorname{csch} u du = \ln \left| \tanh \frac{1}{2} u \right| + C$$

$$109. \int \operatorname{sech}^2 u du = \tanh u + C$$

$$110. \int \operatorname{csch}^2 u du = -\coth u + C$$

$$111. \int \operatorname{sech} u \tanh u du = -\operatorname{sech} u + C$$

$$112. \int \operatorname{csch} u \coth u du = -\operatorname{csch} u + C$$

Forms Involving $\sqrt{2au - u^2}$, $a > 0$

$$113. \int \sqrt{2au - u^2} du = \frac{u - a}{2} \sqrt{2au - u^2} + \frac{a^2}{2} \cos^{-1} \left(\frac{a - u}{a} \right) + C$$

$$114. \int u \sqrt{2au - u^2} du = \frac{2u^2 - au - 3a^2}{6} \sqrt{2au - u^2} + \frac{a^3}{2} \cos^{-1} \left(\frac{a - u}{a} \right) + C$$

$$115. \int \frac{\sqrt{2au - u^2}}{u} du = \sqrt{2au - u^2} + a \cos^{-1} \left(\frac{a - u}{a} \right) + C$$

$$116. \int \frac{\sqrt{2au - u^2}}{u^2} du = -\frac{2\sqrt{2au - u^2}}{u} - \cos^{-1} \left(\frac{a - u}{a} \right) + C$$

$$117. \int \frac{du}{\sqrt{2au - u^2}} = \cos^{-1} \left(\frac{a - u}{a} \right) + C$$

$$118. \int \frac{u du}{\sqrt{2au - u^2}} = -\sqrt{2au - u^2} + a \cos^{-1} \left(\frac{a - u}{a} \right) + C$$

$$119. \int \frac{u^2 du}{\sqrt{2au - u^2}} = -\frac{(u + 3a)}{2} \sqrt{2au - u^2} + \frac{3a^2}{2} \cos^{-1} \left(\frac{a - u}{a} \right) + C$$

$$120. \int \frac{du}{u \sqrt{2au - u^2}} = -\frac{\sqrt{2au - u^2}}{au} + C$$

1. (a) What is a function? What are its domain and range?

A function f is a rule that assigns to each element x in a set D exactly one element, called $f(x)$, in a set E . The set D is called the domain of the function. The range of f is the set of all possible values of $f(x)$ as x varies throughout the domain.

- (b) What is the graph of a function?

If f is a function with domain D , then its graph is the set of ordered pairs $\{(x, f(x)) \mid x \in D\}$.

- (c) How can you tell whether a given curve is the graph of a function?

Use the Vertical Line Test: A curve in the xy -plane is the graph of a function of x if and only if no vertical line intersects the curve more than once.

2. Discuss four ways of representing a function. Illustrate your discussion with examples.

The four ways to represent a function are: verbally, numerically, visually, and algebraically. An example of each is given below.

Verbally: An assignment of students to chairs in a classroom (a description in words)

Numerically: A tax table that assigns an amount of tax to an income (a table of values)

Visually: A graphical history of the Dow Jones average (a graph)

Algebraically: A relationship between distance, speed, and time: $d = rt$ (an explicit formula)

3. (a) What is an even function? How can you tell if a function is even by looking at its graph?

A function f is even if it satisfies $f(-x) = f(x)$ for every number x in its domain. If the graph of a function is symmetric with respect to the y -axis, then f is even. Examples of even functions: $f(x) = x^2$, $f(x) = x^4 + x^2$, $f(x) = |x|$.

- (b) What is an odd function? How can you tell if a function is odd by looking at its graph?

A function f is odd if it satisfies $f(-x) = -f(x)$ for every number x in its domain. If the graph of a function is symmetric with respect to the origin, then f is odd. Examples of odd functions: $f(x) = x^3$, $f(x) = x^3 + x^5$, $f(x) = \sqrt[3]{x}$.

4. What is an increasing function?

A function f is called increasing on an interval I if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ in I .

5. What is a mathematical model?

A mathematical model is a mathematical description (often by means of a function or an equation) of a real-world phenomenon.

6. Give an example of each type of function.

(a) **Linear function:** $f(x) = 2x + 1$, $f(x) = ax + b$

(b) **Power function:** $f(x) = x^2$, $f(x) = x^a$

(c) **Exponential function:** $f(x) = 2^x$, $f(x) = b^x$

(d) **Quadratic function:** $f(x) = x^2 + x + 1$,
 $f(x) = ax^2 + bx + c$ ($a \neq 0$)

(e) **Polynomial of degree 5:** $f(x) = x^5 + 2x^4 - 3x^2 + 7$

(f) **Rational function:** $f(x) = \frac{x}{x+2}$, $f(x) = \frac{P(x)}{Q(x)}$ where $P(x)$ and $Q(x)$ are polynomials

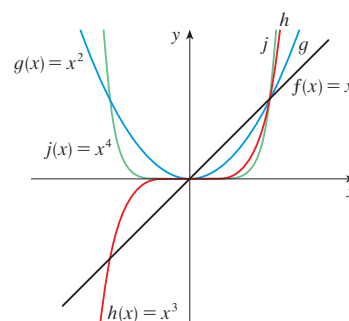
7. Sketch by hand, on the same axes, the graphs of the following functions.

(a) $f(x) = x$

(b) $g(x) = x^2$

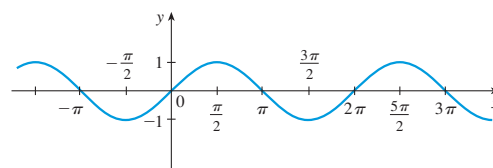
(c) $h(x) = x^3$

(d) $j(x) = x^4$

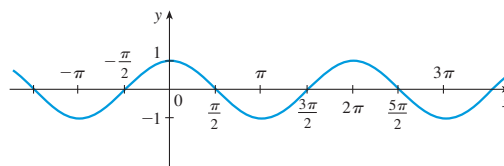


8. Draw, by hand, a rough sketch of the graph of each function.

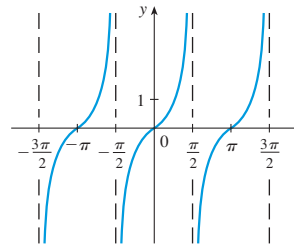
(a) $y = \sin x$



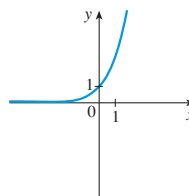
(b) $y = \cos x$



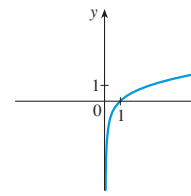
(c) $y = \tan x$



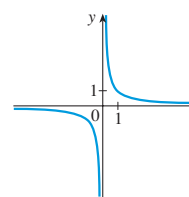
(d) $y = e^x$



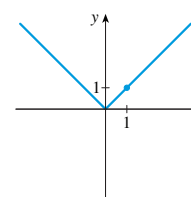
(e) $y = \ln x$



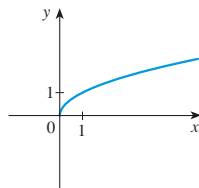
(f) $y = 1/x$



(g) $y = |x|$



(h) $y = \sqrt{x}$



9. Suppose that f has domain A and g has domain B .

(a) What is the domain of $f + g$?

The domain of $f + g$ is the intersection of the domain of f and the domain of g ; that is, $A \cap B$, the numbers that are in both A and B .

(b) What is the domain of fg ?

The domain of fg is also $A \cap B$.

(c) What is the domain of f/g ?

The domain of f/g must exclude values of x that make g equal to 0; that is, $\{x \in A \cap B \mid g(x) \neq 0\}$.

10. How is the composite function $f \circ g$ defined? What is its domain?

Given two functions f and g , the composite function $f \circ g$ is defined by $(f \circ g)(x) = f(g(x))$. The domain of $f \circ g$ is the set of all x in the domain of g such that $g(x)$ is in the domain of f .

11. Suppose the graph of f is given. Write an equation for each of the graphs that are obtained from the graph of f as follows.

(a) Shift 2 units upward: $y = f(x) + 2$

(b) Shift 2 units downward: $y = f(x) - 2$

(c) Shift 2 units to the right: $y = f(x - 2)$

(d) Shift 2 units to the left: $y = f(x + 2)$

(e) Reflect about the x -axis: $y = -f(x)$

(f) Reflect about the y -axis: $y = f(-x)$

(g) Stretch vertically by a factor of 2: $y = 2f(x)$

(h) Shrink vertically by a factor of 2: $y = \frac{1}{2}f(x)$

(i) Stretch horizontally by a factor of 2: $y = f(\frac{1}{2}x)$

(j) Shrink horizontally by a factor of 2: $y = f(2x)$

12. (a) What is a one-to-one function? How can you tell if a function is one-to-one by looking at its graph?

A function f is called a one-to-one function if it never takes on the same value twice; that is, if $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$. (Or, f is 1-1 if each output corresponds to only one input.)

Use the Horizontal Line Test: A function is one-to-one if and only if no horizontal line intersects its graph more than once.

(b) If f is a one-to-one function, how is its inverse function f^{-1} defined? How do you obtain the graph of f^{-1} from the graph of f ?

If f is a one-to-one function with domain A and range B , then its inverse function f^{-1} has domain B and range A and is defined by

$$f^{-1}(y) = x \iff f(x) = y$$

for any y in B . The graph of f^{-1} is obtained by reflecting the graph of f about the line $y = x$.

13. (a) What is a semilog plot?

A semilog plot is a graph of the points $(x, \log y)$ given (x, y) data points.

(b) If a semilog plot of your data lies approximately on a line, what type of model is appropriate?

An exponential model is appropriate when a semilog plot of the data lies on a line.

14. (a) What is a log-log plot?

A log-log plot is a graph of the points $(\log x, \log y)$ given (x, y) data points.

(b) If a log-log plot of your data lies approximately on a line, what type of model is appropriate?

A power model is appropriate when a log-log plot of the data lies on a line.

15. (a) What is a sequence?

A sequence is an ordered list of numbers.

(b) What is a recursive sequence?

A recursive sequence is a sequence in which each term is defined using one or more preceding terms.

16. Discrete-time models

(a) If there are N_t cells at time t and they divide according to the difference equation $N_{t+1} = RN_t$, write an expression for N_t .

$$N_t = N_0 R^t$$

(b) If a population has carrying capacity K , write the logistic difference equation for N_t .

$$N_{t+1} = \left[1 + r \left(1 - \frac{N_t}{K} \right) \right] N_t$$

(c) Write the logistic difference equation for

$$x_t = \frac{r}{(1+r)K} N_t$$

$$x_{t+1} = R_{\max} x_t (1 - x_t)$$

1. (a) What is a convergent sequence?

A convergent sequence is an ordered list of numbers $\{a_n\}$ for which $\lim_{n \rightarrow \infty} a_n$ exists.

(b) What does $\lim_{n \rightarrow \infty} a_n = 3$ mean?

The terms of the sequence $\{a_n\}$ approach 3 as n becomes large.

2. What is $\lim_{n \rightarrow \infty} r^n$ in the following three cases?

(a) $0 < r < 1$

$$\lim_{n \rightarrow \infty} r^n = 0 \text{ when } 0 < r < 1$$

(b) $r = 1$

$$\lim_{n \rightarrow \infty} r^n = 1 \text{ when } r = 1$$

(c) $r > 1$

$$\lim_{n \rightarrow \infty} r^n = \infty \text{ when } r > 1$$

3. (a) What is the sum of the finite geometric series

$$a + ar + ar^2 + \cdots + ar^n$$

$$a + ar + ar^2 + \cdots + ar^n = \frac{a(1 - r^{n+1})}{1 - r}$$

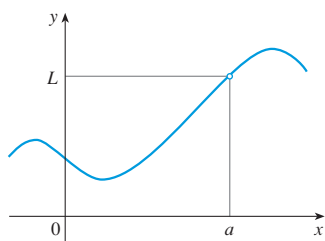
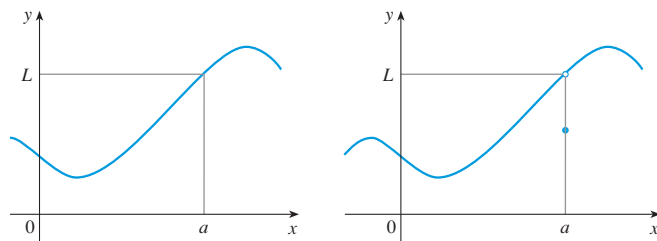
(b) If $-1 < r < 1$, what is the sum of the infinite geometric series $a + ar + ar^2 + \cdots + ar^n + \cdots$?

$$a + ar + ar^2 + \cdots + ar^n + \cdots = \frac{a}{1 - r} \text{ if } -1 < r < 1$$

4. Explain what each of the following means and illustrate with a sketch.

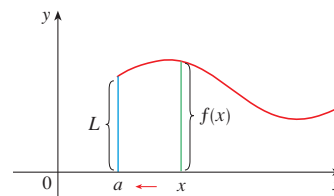
(a) $\lim_{x \rightarrow a} f(x) = L$

$\lim_{x \rightarrow a} f(x) = L$ means that the values of $f(x)$ approach L as the values of x approach a (from either side).



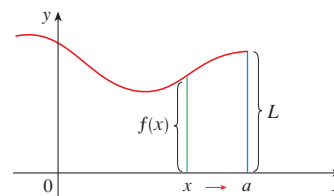
(b) $\lim_{x \rightarrow a^+} f(x) = L$

$\lim_{x \rightarrow a^+} f(x) = L$ means that the values of $f(x)$ approach L as the values of x approach a from the right side.



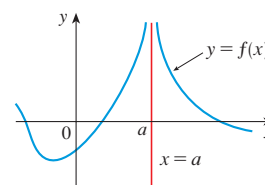
(c) $\lim_{x \rightarrow a^-} f(x) = L$

$\lim_{x \rightarrow a^-} f(x) = L$ means that the values of $f(x)$ approach L as the values of x approach a from the left side.



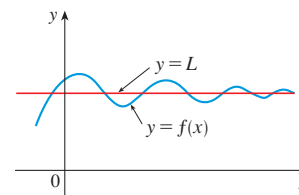
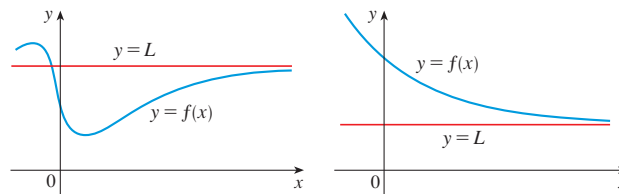
(d) $\lim_{x \rightarrow a} f(x) = \infty$

$\lim_{x \rightarrow a} f(x) = \infty$ means that the values of $f(x)$ become arbitrarily large as x approaches a (from either side).



(e) $\lim_{x \rightarrow \infty} f(x) = L$

This means that the values of $f(x)$ approach L as x becomes large.



5. State the following Limit Laws for functions.

(a) Sum Law

The limit of a sum is the sum of the limits:

$$\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

(continued)

(b) **Difference Law**

The limit of a difference is the difference of the limits:

$$\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

(c) **Constant Multiple Law**

The limit of a constant times a function is the constant times the limit of the function: $\lim_{x \rightarrow a} [cf(x)] = c \lim_{x \rightarrow a} f(x)$

(d) **Product Law**

The limit of a product is the product of the limits:

$$\lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

(e) **Quotient Law**

The limit of a quotient is the quotient of the limits, provided that the limit of the denominator is not 0:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \quad \text{if } \lim_{x \rightarrow a} g(x) \neq 0$$

(f) **Power Law**

The limit of a power is the power of the limit:

$$\lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n \quad (\text{for } n \text{ a positive integer})$$

(g) **Root Law**

The limit of a root is the root of the limit:

$$\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)} \quad \text{where } n \text{ is a positive integer}$$

[If n is even, assume that $\lim_{x \rightarrow a} f(x) > 0$.]

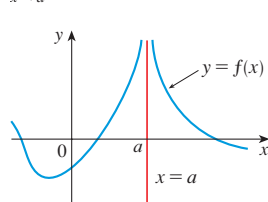
6. What does the Squeeze Theorem say?

The Squeeze Theorem states that if $f(x) \leq g(x) \leq h(x)$ when x is near a (except possibly at a) and $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$, then $\lim_{x \rightarrow a} g(x) = L$.

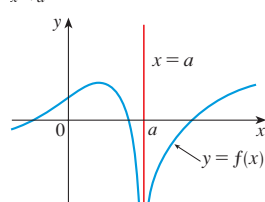
7. (a) What does it mean to say that the line $x = a$ is a vertical asymptote of the curve $y = f(x)$? Draw curves to illustrate the various possibilities.

The line $x = a$ is a vertical asymptote of the curve $y = f(x)$ if at least one of the following is true:

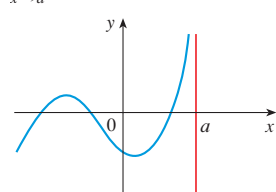
$$\lim_{x \rightarrow a} f(x) = \infty$$



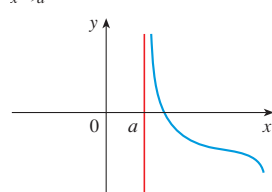
$$\lim_{x \rightarrow a} f(x) = -\infty$$



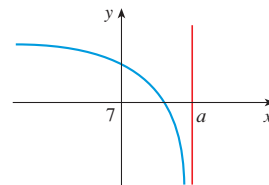
$$\lim_{x \rightarrow a^-} f(x) = \infty$$



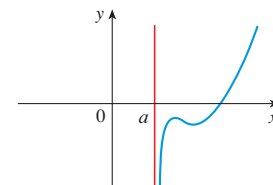
$$\lim_{x \rightarrow a^+} f(x) = \infty$$



$$\lim_{x \rightarrow a^-} f(x) = -\infty$$

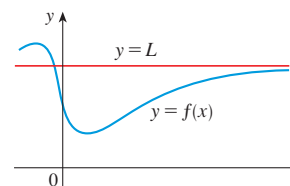
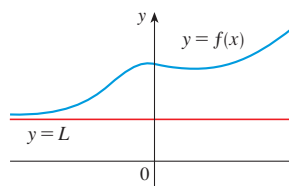


$$\lim_{x \rightarrow a^+} f(x) = -\infty$$



(b) What does it mean to say that the line $y = L$ is a horizontal asymptote of the curve $y = f(x)$? Draw curves to illustrate the various possibilities.

The line $y = L$ is a horizontal asymptote of the curve $y = f(x)$ if either $\lim_{x \rightarrow \infty} f(x) = L$ or $\lim_{x \rightarrow -\infty} f(x) = L$.



8. Which of the following curves have vertical asymptotes? Which have horizontal asymptotes?

(a) $y = x^4$: No asymptote

(b) $y = \sin x$: No asymptote

(c) $y = \tan x$: Vertical asymptotes $x = \frac{\pi}{2} + \pi n$, n an integer

(d) $y = e^x$: Horizontal asymptote $y = 0$ ($\lim_{x \rightarrow -\infty} e^x = 0$)

(e) $y = \ln x$: Vertical asymptote $x = 0$ ($\lim_{x \rightarrow 0^+} \ln x = -\infty$)

(f) $y = 1/x$:

Vertical asymptote $x = 0$, horizontal asymptote $y = 0$

(g) $y = \sqrt{x}$: No asymptote

9. (a) What does it mean for f to be continuous at a ?

A function f is continuous at a number a if $f(x)$ approaches $f(a)$ as x approaches a ; that is, $\lim_{x \rightarrow a} f(x) = f(a)$.

(b) What does it mean for f to be continuous on the interval $(-\infty, \infty)$? What can you say about the graph of such a function?

A function f is continuous on the interval $(-\infty, \infty)$ if f is continuous at every real number a . The graph of such a function has no break and every vertical line crosses it.

10. What does the Intermediate Value Theorem say?

The Intermediate Value Theorem states that if f is continuous on the closed interval $[a, b]$ and N is any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$, then there exists a number c in (a, b) such that $f(c) = N$. In other words, a continuous function takes on every intermediate value between the function values $f(a)$ and $f(b)$.

1. Write an expression for the slope of the tangent line to the curve $y = f(x)$ at the point $(a, f(a))$.

The slope of the tangent line to the graph of $y = f(x)$ at the point $(a, f(a))$ is given by

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \quad \text{or} \quad \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

2. Define the derivative $f'(a)$. Discuss two ways of interpreting this number.

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h} \quad \text{or, equivalently,}$$

$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$. The derivative $f'(a)$ is the instantaneous rate of change of f (with respect to x) when $x = a$ and also represents the slope of the tangent line to the graph of f at $x = a$.

3. If $y = f(x)$ and x changes from x_1 to x_2 , write expressions for the following.

- (a) The average rate of change of y with respect to x over the interval $[x_1, x_2]$.

The average rate of change of y with respect to x over the interval $[x_1, x_2]$ is $\frac{f(x_2) - f(x_1)}{x_2 - x_1}$.

- (b) The instantaneous rate of change of y with respect to x at $x = x_1$.

The instantaneous rate of change of y with respect to x at $x = x_1$ is $\lim_{x_2 \rightarrow x_1} \frac{f(x_2) - f(x_1)}{x_2 - x_1}$.

4. Define the second derivative of f . If $f(t)$ is the position function of a particle, how can you interpret the second derivative?

The second derivative of f , f'' , is the derivative of f' . If $f(t)$ is the position function of a particle, then the first derivative is velocity and the second derivative is the derivative of velocity, namely, acceleration.

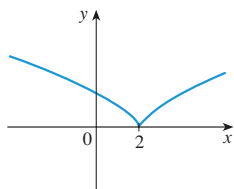
5. (a) What does it mean for f to be differentiable at a ?

A function f is differentiable at a number a if its derivative f' exists at $x = a$; that is, if $f'(a)$ exists.

- (b) What is the relation between the differentiability and continuity of a function?

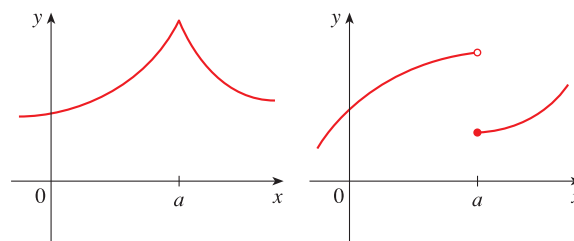
If f is differentiable at a , then f is continuous at a . So if f is not continuous at a , then f is not differentiable at a .

- (c) Sketch the graph of a function that is continuous but not differentiable at $a = 2$.



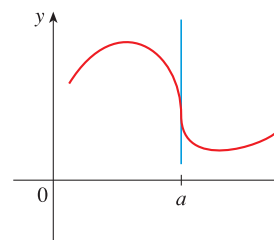
6. Describe several ways in which a function can fail to be differentiable. Illustrate with sketches.

A function is not differentiable at any value where the graph has a corner, where the graph has a discontinuity, or where it has a vertical tangent line.



A corner

A discontinuity



A vertical tangent

7. State each differentiation rule both in symbols and in words.

- (a) The Power Rule

If n is any real number, then $\frac{d}{dx}(x^n) = nx^{n-1}$. The derivative of a variable base raised to a constant power is the power times the base raised to the power minus one.

- (b) The Constant Multiple Rule

If c is a constant and f is a differentiable function, then $\frac{d}{dx}[cf(x)] = c \frac{d}{dx}f(x)$. The derivative of a constant times a function is the constant times the derivative of the function.

- (c) The Sum Rule

If f and g are both differentiable, then $\frac{d}{dx}[f(x) + g(x)] = \frac{d}{dx}f(x) + \frac{d}{dx}g(x)$. The derivative of a sum of functions is the sum of the derivatives.

- (d) The Difference Rule

If f and g are both differentiable, then $\frac{d}{dx}[f(x) - g(x)] = \frac{d}{dx}f(x) - \frac{d}{dx}g(x)$. The derivative of a difference of functions is the difference of the derivatives.

- (e) The Product Rule

If f and g are both differentiable, then $\frac{d}{dx}[f(x)g(x)] = f(x) \frac{d}{dx}g(x) + g(x) \frac{d}{dx}f(x)$

The derivative of a product of two functions is the first function times the derivative of the second function plus the second function times the derivative of the first function.

(f) The Quotient Rule

If f and g are both differentiable, then

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x) \frac{d}{dx} f(x) - f(x) \frac{d}{dx} g(x)}{[g(x)]^2}$$

The derivative of a quotient of functions is the denominator times the derivative of the numerator minus the numerator times the derivative of the denominator, all divided by the square of the denominator.

(g) The Chain Rule

If g is differentiable at x and f is differentiable at $g(x)$, then the composite function defined by $F(x) = f(g(x))$ is differentiable at x and F' is given by the product $F'(x) = f'(g(x))g'(x)$. The derivative of a composite function is the derivative of the outer function evaluated at the inner function times the derivative of the inner function.

8. State the derivative of each function.

- (a) $y = x^n$: $y' = nx^{n-1}$
- (b) $y = e^x$: $y' = e^x$
- (c) $y = b^x$: $y' = b^x \ln b$
- (d) $y = \ln x$: $y' = 1/x$
- (e) $y = \log_b x$: $y' = 1/(x \ln b)$
- (f) $y = \sin x$: $y' = \cos x$
- (g) $y = \cos x$: $y' = -\sin x$
- (h) $y = \tan x$: $y' = \sec^2 x$
- (i) $y = \csc x$: $y' = -\csc x \cot x$
- (j) $y = \sec x$: $y' = \sec x \tan x$
- (k) $y = \cot x$: $y' = -\csc^2 x$
- (l) $y = \tan^{-1} x$: $y' = 1/(1 + x^2)$

9. (a) How is the number e defined?

e is the number such that $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$.

(b) Express e as a limit.

$$e = \lim_{x \rightarrow 0} (1 + x)^{1/x}$$

(c) Why is the natural exponential function $y = e^x$ used more often in calculus than the other exponential functions $y = b^x$?

The differentiation formula for $y = b^x$ [$y' = b^x \ln b$] is simplest when $b = e$ because $\ln e = 1$.

(d) Why is the natural logarithmic function $y = \ln x$ used more often in calculus than the other logarithmic functions $y = \log_b x$?

The differentiation formula for $y = \log_b x$ [$y' = 1/(x \ln b)$] is simplest when $b = e$ because $\ln e = 1$.

10. (a) Explain how implicit differentiation works. When should you use it?

Implicit differentiation consists of differentiating both sides of an equation involving x and y with respect to x , and then solving the resulting equation for y' . Use implicit differentiation when it is difficult to solve an equation for y in terms of x .

(b) Explain how logarithmic differentiation works. When should you use it?

Logarithmic differentiation consists of taking natural logarithms of both sides of an equation $y = f(x)$, simplifying, differentiating implicitly with respect to x , and then solving the resulting equation for y' . Use logarithmic differentiation when the calculation of derivatives of complicated functions involving products, quotients, or powers can be simplified by taking logarithms.

11. Write an expression for the linearization of f at a .

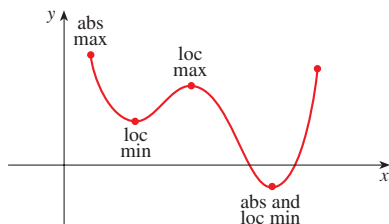
$$L(x) = f(a) + f'(a)(x - a).$$

12. Write an expression for the n th-degree Taylor polynomial of f centered at a .

$$T_n(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n$$

1. Explain the difference between an absolute maximum and a local maximum. Illustrate with a sketch.

The function value $f(c)$ is the absolute maximum value of f if $f(c)$ is the largest function value on the entire domain of f , whereas f is a local maximum value if it is the largest function value when x is near c .



2. (a) What does the Extreme Value Theorem say?

The Extreme Value Theorem states that if f is a continuous function on a closed interval $[a, b]$, then it always attains an absolute maximum and an absolute minimum value on that interval.

- (b) Explain how the Closed Interval Method works.

To find the absolute maximum and minimum values of a continuous function f on a closed interval $[a, b]$, we follow these three steps:

- Find the critical numbers of f in the interval (a, b) and compute the values of f at these numbers.
- Find the values of f at the endpoints of the interval.
- The largest of the output values from the previous two steps is the absolute maximum value; the smallest of these values is the absolute minimum value.

3. (a) State Fermat's Theorem.

Fermat's Theorem states that if f has a local maximum or minimum at c , and if $f'(c)$ exists, then $f'(c) = 0$.

- (b) Define a critical number of f .

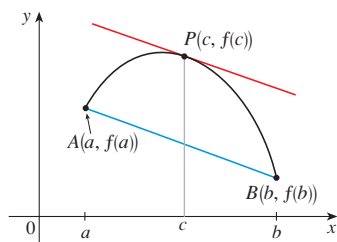
A critical number of a function f is a number c in the domain of f such that either $f'(c) = 0$ or $f'(c)$ does not exist.

4. State the Mean Value Theorem and give a geometric interpretation.

The Mean Value Theorem states that if f is a differentiable function on the interval $[a, b]$, then there exists a number c between a

and b such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Geometric interpretation: There is some point P on the graph of a function f [on the interval (a, b)] where the tangent line is parallel to the secant line that connects $(a, f(a))$ and $(b, f(b))$.



5. (a) State the Increasing/Decreasing Test.

If $f'(x) > 0$ on an interval, then f is increasing on that interval.

If $f'(x) < 0$ on an interval, then f is decreasing on that interval.

- (b) What does it mean to say that f is concave upward on an interval I ?

A function f is concave upward on an interval I if f' is an increasing function on I (or, equivalently, the graph of f lies above all of its tangent lines on I).

- (c) State the Concavity Test.

If $f''(x) > 0$ on an interval, then the graph of f is concave upward on that interval.

If $f''(x) < 0$ on an interval, then the graph of f is concave downward on that interval.

- (d) What are inflection points? How do you find them?

Inflection points on the graph of a continuous function f are points where the curve changes from concave upward to concave downward or from concave downward to concave upward. They can be found by determining the values at which the second derivative changes sign.

6. (a) State the First Derivative Test.

Suppose that c is a critical number of a continuous function f .

- If f' changes from positive to negative at c , then f has a local maximum at c .
- If f' changes from negative to positive at c , then f has a local minimum at c .
- If f' does not change sign at c , then f has no local maximum or minimum at c .

- (b) State the Second Derivative Test.

Suppose f'' is continuous near c .

- If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c .
- If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c .

- (c) What are the relative advantages and disadvantages of these tests?

The Second Derivative Test is sometimes easier to use, but it is inconclusive when $f''(c) = 0$ and fails if $f''(c)$ does not exist. In either case the First Derivative Test must be used.

7. (a) What does l'Hospital's Rule say?

L'Hospital's Rule states that if f and g are differentiable functions, $g'(x) \neq 0$ near a (except possibly at a), and $\lim_{x \rightarrow a} f(x)/g(x)$ is an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then

$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ provided the right side limit exists.

(continued)

- (b) How can you use l'Hospital's Rule if you have a product $f(x)g(x)$ where $f(x) \rightarrow 0$ and $g(x) \rightarrow \infty$ as $x \rightarrow a$?

Write fg as $\frac{f}{1/g}$ or $\frac{g}{1/f}$.

- (c) How can you use l'Hospital's Rule if you have a difference $f(x) - g(x)$ where $f(x) \rightarrow \infty$ and $g(x) \rightarrow \infty$ as $x \rightarrow a$?

Convert the difference into a quotient using a common denominator, rationalizing, factoring, or some other method.

8. (a) What is an equilibrium of the recursive sequence $x_{t+1} = f(x_t)$?

An equilibrium of the recursive sequence $x_{t+1} = f(x_t)$ is a number $\hat{x}$ that is left unchanged by the function f , that is, $f(\hat{x}) = \hat{x}$.

- (b) What is a stable equilibrium? An unstable equilibrium?

An equilibrium is stable if solutions that begin close to the equilibrium approach that equilibrium. An equilibrium is

unstable if solutions that start close to the equilibrium move away from it.

- (c) State the Stability Criterion.

If $\hat{x}$ is an equilibrium of the recursive sequence $x_{t+1} = f(x_t)$, where f' is continuous, and $|f'(\hat{x})| < 1$, then the equilibrium is stable. If $|f'(\hat{x})| > 1$, then the equilibrium is unstable.

9. (a) What is an antiderivative of a function f ?

A function F is an antiderivative of f on an interval I if $F'(x) = f(x)$ for all x in I .

- (b) Suppose F_1 and F_2 are both antiderivatives of f on an interval I . How are F_1 and F_2 related?

If F_1 and F_2 are both antiderivatives of f on an interval I , then they differ by a constant.

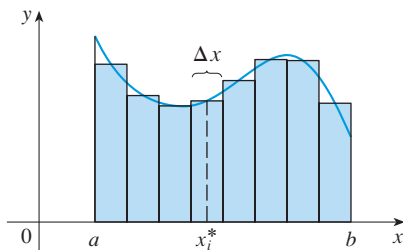
1. (a) Write an expression for a Riemann sum of a function f .

Explain the meaning of the notation that you use.

$\sum_{i=1}^n f(x_i^*) \Delta x$ is an expression for a Riemann sum of a function f , where x_i^* is a point in the i th subinterval $[x_{i-1}, x_i]$ and Δx is the length of the subintervals.

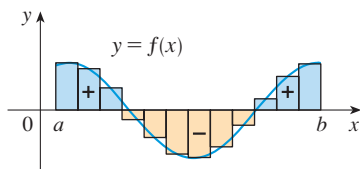
- (b) If $f(x) \geq 0$, what is the geometric interpretation of a Riemann sum? Illustrate with a diagram.

If f is positive, then a Riemann sum can be interpreted as the sum of areas of approximating rectangles, as shown in the figure.



- (c) If $f(x)$ takes on both positive and negative values, what is the geometric interpretation of a Riemann sum? Illustrate with a diagram.

If f takes on both positive and negative values, as in the figure below, then the Riemann sum $\sum f(x_i^*) \Delta x$ is the sum of the areas of the rectangles that lie above the x -axis and the negatives of the areas of the rectangles that lie below the x -axis (the areas of the blue rectangles minus the areas of the gold rectangles).



2. (a) Write the definition of the definite integral of a continuous function from a to b .

If f is a function defined for $a \leq x \leq b$, we divide the interval $[a, b]$ into n subintervals of equal width $\Delta x = (b - a)/n$. We let $x_0 (= a), x_1, x_2, \dots, x_n (= b)$ be the endpoints of these subintervals and we let $x_1^*, x_2^*, \dots, x_n^*$ be any sample points in these subintervals, so x_i^* lies in the i th subinterval $[x_{i-1}, x_i]$. Then the definite integral of f from a to b is

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

provided that this limit exists.

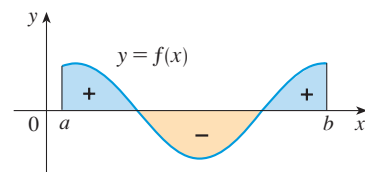
- (b) What is the geometric interpretation of $\int_a^b f(x) dx$ if $f(x) \geq 0$?

If f is positive, then $\int_a^b f(x) dx$ can be interpreted as the area under the graph of $y = f(x)$ and above the x -axis for $a \leq x \leq b$.

- (c) What is the geometric interpretation of $\int_a^b f(x) dx$ if $f(x)$ takes on both positive and negative values? Illustrate with a diagram.

In this case $\int_a^b f(x) dx$ can be interpreted as a “net area,” that is, the area of the region above the x -axis and below the graph

of f (labeled “+” in the figure) minus the area of the region below the x -axis and above the graph of f (labeled “−”).



3. State the Midpoint Rule.

If f is a continuous function on the interval $[a, b]$ and we divide $[a, b]$ into n subintervals of equal width $\Delta x = (b - a)/n$, then

$$\int_a^b f(x) dx \approx \sum_{i=1}^n f(\bar{x}_i) \Delta x = \Delta x [f(\bar{x}_1) + \dots + f(\bar{x}_n)]$$

where $\bar{x}_i = \frac{1}{2}(x_{i-1} + x_i) =$ midpoint of $[x_{i-1}, x_i]$.

4. (a) State the Evaluation Theorem.

The Evaluation Theorem says that if f is continuous on the interval $[a, b]$, then $\int_a^b f(x) dx = F(b) - F(a)$, where F is any antiderivative of f , that is, $F' = f$.

- (b) State the Net Change Theorem.

The Net Change Theorem says that the integral of a rate of change is the net change: $\int_a^b F'(x) dx = F(b) - F(a)$.

5. If $r(t)$ is the rate of growth of a population at time t , where t is measured in months, what does $\int_6^{10} r(t) dt$ represent?

$\int_6^{10} r(t) dt$ is the change in population size from month 6 to month 10.

6. (a) Explain the meaning of the indefinite integral $\int f(x) dx$.

The indefinite integral $\int f(x) dx$ is another name for an antiderivative of f , so $\int f(x) dx = F(x)$ means that $F'(x) = f(x)$.

- (b) What is the connection between the definite integral $\int_a^b f(x) dx$ and the indefinite integral $\int f(x) dx$?

The connection is given by the Fundamental Theorem:

$$\int_a^b f(x) dx = \int f(x) dx \Big|_a^b$$

if f is continuous.

7. State both parts of the Fundamental Theorem of Calculus.

Suppose f is continuous on $[a, b]$. The Fundamental Theorem of Calculus says

1. If $g(x) = \int_a^x f(t) dt$, then $g'(x) = f(x)$.

2. $\int_a^b f(x) dx = F(b) - F(a)$, where F is any antiderivative of f , that is, $F' = f$.

8. (a) State the Substitution Rule. In practice, how do you use it?

The substitution rule says that if $u = g(x)$ is a differentiable function whose range is an interval I and f is continuous on I , then $\int f(g(x)) g'(x) dx = \int f(u) du$. In practice, we make the substitutions $u = g(x)$ and $du = g'(x) dx$ in the integrand to make the integral simpler to evaluate.

(continued)

- (b) State the rule for integration by parts. In practice, how do you use it?

The rule for integration by parts states that $\int f(x)g'(x) dx = f(x)g(x) - \int g(x)f'(x) dx$. In practice, we try to choose $u = f(x)$ to be a function that becomes simpler when differentiated (or at least not more complicated) as long as $dv = g'(x) dx$ can be readily integrated to give v . Then the original integral $\int u dv$ becomes $uv - \int v du$.

9. Define the following improper integrals.

(a) $\int_a^\infty f(x) dx$

If $\int_a^b f(x) dx$ exists for every number $b \geq a$, then

$\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx$ provided this limit exists (as a finite number).

(b) $\int_{-\infty}^b f(x) dx$

If $\int_a^b f(x) dx$ exists for every number $a \leq b$, then

$\int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx$ provided this limit exists (as a finite number).

(c) $\int_{-\infty}^\infty f(x) dx$

If both $\int_c^\infty f(x) dx$ and $\int_{-\infty}^c f(x) dx$ are convergent, then we define $\int_{-\infty}^\infty f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^\infty f(x) dx$, where c is any real number.

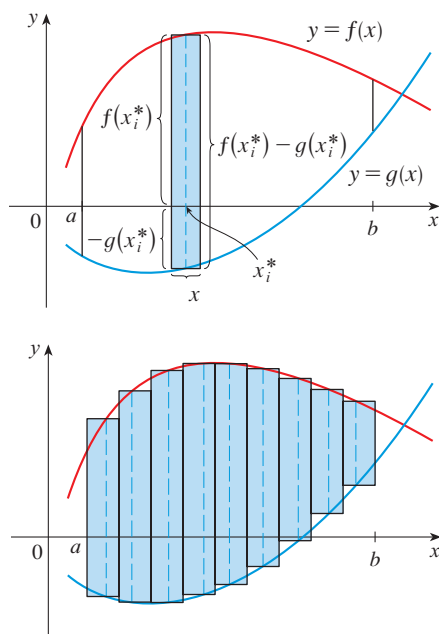
10. Explain exactly what is meant by the statement that “differentiation and integration are inverse processes.”

The Fundamental Theorem of Calculus (or, equivalently, the Net Change Theorem) states that $\int_a^b F'(x) dx = F(b) - F(a)$. This says that if we take a function F , first differentiate it, and then integrate the result, we arrive back at the original function, but in the form $F(b) - F(a)$. Also, the indefinite integral $\int f(x) dx$ represents an antiderivative of f , so $\frac{d}{dx} \int f(x) dx = f(x)$. Here we first integrate a function, then differentiate the result, and arrive back at the original function.

1. Draw two typical curves $y = f(x)$ and $y = g(x)$, where $f(x) \geq g(x)$ for $a \leq x \leq b$. Show how to approximate the area between these curves by a Riemann sum and sketch the corresponding approximating rectangles. Then write an expression for the exact area.

If we divide the interval $[a, b]$ into n subintervals of equal width Δx and label the endpoints of the subintervals $x_0, x_1, x_2, \dots, x_n$, then the area between $f(x)$ and $g(x)$ can be approximated by the Riemann sum $\sum_{i=1}^n [f(x_i^*) - g(x_i^*)] \Delta x$, where x_i^* lies in the interval $[x_{i-1}, x_i]$. In the limit as $n \rightarrow \infty$ the Riemann sum approximation gives the exact area

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n [f(x_i^*) - g(x_i^*)] \Delta x = \int_a^b [f(x) - g(x)] dx$$



2. Suppose that Sue runs faster than Kathy throughout a 1500-meter race. What is the physical meaning of the area between their velocity curves for the first minute of the race?

It represents the number of meters by which Sue is ahead of Kathy after 1 minute.

3. (a) What is the average value of a function f on an interval $[a, b]$?

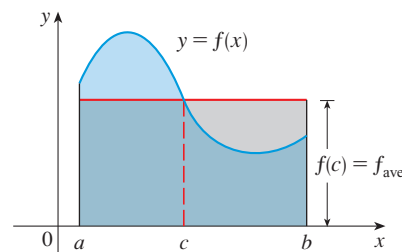
The average value of a function f on an interval $[a, b]$ is

$$f_{\text{ave}} = \frac{1}{b-a} \int_a^b f(x) dx.$$

- (b) What does the Mean Value Theorem for Integrals say? What is its geometric interpretation?

The Mean Value Theorem for Integrals says that if f is continuous on $[a, b]$, then there exists a number c in $[a, b]$ such that $\int_a^b f(x) dx = f(c)(b-a)$. The geometric interpretation is that, for positive functions f , there is a number c such that the

rectangle with base $[a, b]$ and height $f(c)$ has the same area as the region under the graph of f from a to b .



4. If we have survival and renewal functions for a population, how do we predict the size of the population T years from now?

If a population begins with P_0 members, new members are added at a rate given by the renewal function $R(t)$, where t is measured in years, and the proportion of the population that remains after t years is given by the survival function $S(t)$, then the population T years from now is given by

$$P(T) = S(T) \cdot P_0 + \int_0^T S(T-t) R(t) dt$$

5. (a) What is the cardiac output of the heart?

The cardiac output of the heart is the volume of blood pumped by the heart per unit time.

- (b) Explain how the cardiac output can be measured by the dye dilution method.

Dye is injected into part of the heart and a probe measures the concentration of the dye leaving the heart over a time interval $[0, T]$ until the dye has cleared. If $c(t)$ is the concentration of the dye at time t , then the cardiac output is given by

$$F = \frac{A}{\int_0^T c(t) dt}$$

where A is the amount of dye used.

6. (a) Suppose S is a solid with known cross-sectional areas. Explain how to approximate the volume of S by a Riemann sum. Then write an expression for the exact volume.

If we divide S into n “slabs” of equal width Δx , label the endpoints of the slabs $x_0, x_1, x_2, \dots, x_n$, and the cross-sectional area $A(x)$ of the solid is known, then the volume of S can be approximated by the Riemann sum $\sum_{i=1}^n A(x_i^*) \Delta x$, where x_i^* lies in the interval $[x_{i-1}, x_i]$. In the limit as $n \rightarrow \infty$ the Riemann sum approximation gives the exact volume

$$V = \lim_{n \rightarrow \infty} \sum_{i=1}^n A(x_i^*) \Delta x = \int_a^b A(x) dx$$

- (b) If S is a solid of revolution, how do you find the cross-sectional areas?

If the cross-section is a disk, we find the radius in terms of x or y and use $A = \pi(\text{radius})^2$. If the cross-section is a washer, we find the inner radius r_{in} and outer radius r_{out} and use $A = \pi(r_{\text{out}}^2) - \pi(r_{\text{in}}^2)$.

1. (a) What is a differential equation?

A differential equation is an equation that contains an unknown function and one or more of its derivatives.

(b) What is the order of a differential equation?

The order of a differential equation is the order of the highest derivative that occurs in the equation.

(c) What is an initial condition?

An initial condition is a condition of the form $y(t_0) = y_0$.

(d) What are the differences between pure-time, autonomous, and nonautonomous differential equations?

Pure-time differential equations involve the derivative of the function but not the function itself and can be expressed in the form

$$\frac{dy}{dt} = f(t)$$

Autonomous differential equations involve both the derivative of the function and the function itself, but have no explicit dependence on the independent variable. They can be written as

$$\frac{dy}{dt} = g(y)$$

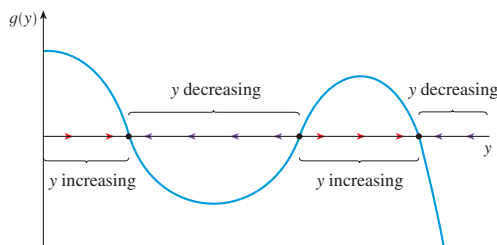
Nonautonomous differential equations are a combination of pure-time and autonomous differential equations. They involve the function, its derivative, and the independent variable explicitly.

2. What can you say about the solutions of the equation $y' = x^2 + y^2$ just by looking at the differential equation?

$y' = x^2 + y^2 \geq 0$ for all x and y . Also, $y' = 0$ only at the origin, so there is a horizontal tangent at $(0, 0)$, but nowhere else. The graph of the solution is increasing on every interval.

3. What is a phase plot for the differential equation $y' = g(y)$?

A phase plot helps to visualize the equilibria of $y' = g(y)$ and to determine the stability properties of the equilibria. A phase plot is constructed by graphing $g(y)$ as a function of y and drawing arrows on the horizontal axis to indicate the direction of change of y . Arrows point to the right when the curve lies above the horizontal axis and to the left when the curve lies below the horizontal axis. The points where the curve intersects the horizontal axis are the equilibrium points.

4. What is a direction field for the differential equation $y' = F(x, y)$?

A direction field is a plot that helps visualize the general shape of the solution curves for the differential equation $y' = F(x, y)$. The direction field can be constructed by drawing short line segments with slope $F(x, y)$ at several points (x, y) . These line segments indicate the direction in which a solution curve is moving.

5. Explain how Euler's method works.

Euler's method is used to approximate values for the solution of the initial-value problem $y' = F(t, y)$, $y(t_0) = t_0$. The method involves starting at the initial point (t_0, y_0) , moving a short distance in the direction indicated by the direction field, stopping and proceeding in the new direction given by the direction field, then repeat. More formally, if the step size is h , the approximate values at $t_{n+1} = t_n + h$ are $y_{n+1} = y_n + hF(t_n, y_n)$.

6. What is a separable differential equation? How do you solve it?

It is a first-order differential equation in which the expression for dy/dx can be factored as a function of x times a function of y , that is, $dy/dx = g(x)f(y)$. We can solve the equation by integrating both sides of the equation $dy/f(y) = g(x) dx$ and solving for y .

7. (a) Write the logistic equation.

$$dN/dt = rN \left(1 - \frac{N}{K} \right), \text{ where } K \text{ is the carrying capacity.}$$

(b) Under what circumstances is this an appropriate model for population growth?

It is an appropriate model for population growth if the population grows at a rate proportional to the size of the population in the beginning, but eventually levels off and approaches its carrying capacity because of limited resources.

8. (a) Write Lotka-Volterra equations to model populations of sharks S and their food F .

$$\frac{dF}{dt} = rF - aFS \quad \frac{dS}{dt} = -kS + bFS$$

(b) What do these equations say about each population in the absence of the other?

In the absence of sharks, the food grows exponentially, that is, $dF/dt = rF$, where r is a positive constant. In the absence of food, the shark population declines at a rate proportional to itself, that is, $dS/dt = -kS$, where k is a positive constant.

9. What is a nullcline?

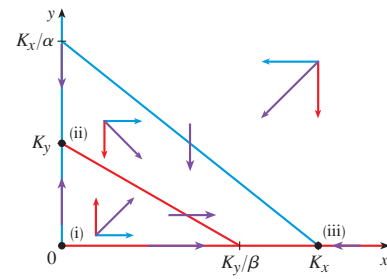
A nullcline of a specific variable is a curve along which that variable does not change. For example, if $dx/dt = f(x, y)$ then the x -nullclines are the curves in the xy -plane that satisfy $f(x, y) = 0$. Along these curves, $dx/dt = 0$.

10. (a) Write Lotka-Volterra competition equations for two competing fish species, x and y .

$$\frac{dx}{dt} = xr \left(1 - \frac{x + \alpha y}{K_x} \right) \quad \frac{dy}{dt} = yr \left(1 - \frac{y + \beta x}{K_y} \right)$$

- (b) What would the nullclines have to look like for species x to always outcompete species y ?

The nullclines must result in movement away from equilibria (i) and (ii), but toward equilibrium (iii), as seen in the phase plane at the right.

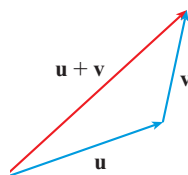


1. What is the difference between a vector and a scalar?

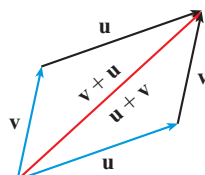
A scalar is a real number, whereas a vector is a quantity that has both a magnitude and a direction.

2. How do you add two vectors geometrically? How do you add them algebraically?

To add two vectors geometrically, we can use either the Triangle Law or the Parallelogram Law, as illustrated in the figures below.



The Triangle Law



The Parallelogram Law

Algebraically, we add the corresponding components of the vectors.

3. If $\mathbf{a}$ is a vector and c is a scalar, how is $c\mathbf{a}$ related to $\mathbf{a}$ geometrically? How do you find $c\mathbf{a}$ algebraically?

$c\mathbf{a}$ is the vector whose length is $|c|$ times the length of $\mathbf{a}$ and whose direction is the same as $\mathbf{a}$ if $c > 0$ and is opposite to $\mathbf{a}$ if $c < 0$. If $c = 0$ or $\mathbf{a} = \mathbf{0}$, then $c\mathbf{a} = \mathbf{0}$. Algebraically, to find $c\mathbf{a}$ we multiply each component of $\mathbf{a}$ by c .

4. How do you find the vector from one point to another algebraically?

The vector from point $A(x_1, y_1, z_1)$ to point $B(x_2, y_2, z_2)$ is given by $\vec{AB} = [x_2 - x_1, y_2 - y_1, z_2 - z_1]$.

5. How do you find the dot product $\mathbf{a} \cdot \mathbf{b}$ of two vectors if you know their lengths and the angle between them? What if you know their components?

When the lengths of $\mathbf{a}$ and $\mathbf{b}$ and the angle θ between them are known, the dot product can be computed using

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta$$

If the components of $\mathbf{a} = [a_1, a_2, a_3]$ and $\mathbf{b} = [b_1, b_2, b_3]$ are known, then the dot product is given by

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

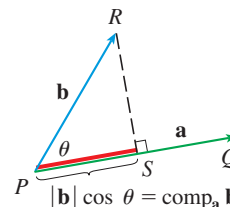
That is, we multiply corresponding components and add.

6. How are dot products useful?

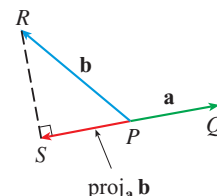
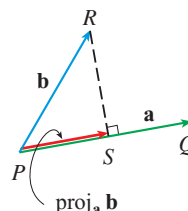
The dot product can be used to find the angle between two vectors, to calculate the scalar or vector projection of one vector onto another, and to write equations of planes. In particular, the dot product can determine if two vectors are orthogonal. Also, the dot product can be used to analyze genome expression profiles for biological discovery and to diagnose cardiac problems using vectorcardiography.

7. Write expressions for the scalar and vector projections of $\mathbf{b}$ onto $\mathbf{a}$. Illustrate with diagrams.

Scalar projection of $\mathbf{b}$ onto $\mathbf{a}$: $\text{comp}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|}$



Vector projection of $\mathbf{b}$ onto $\mathbf{a}$: $\text{proj}_{\mathbf{a}} \mathbf{b} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|^2} \right) \mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|^2} \mathbf{a}$



8. What is the equation of a sphere?

An equation of a sphere with center $C(h, k, l)$ and radius r is $(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$.

9. (a) How do you tell if two vectors are parallel?

Two (nonzero) vectors are parallel if and only if one is a scalar multiple of the other. In addition, two nonzero vectors are parallel if and only if their dot product is the product of their lengths.

(b) How do you tell if two vectors are perpendicular?

Two vectors are perpendicular if and only if their dot product is 0.

10. What is a symmetric matrix?

A matrix A is called symmetric if $A^T = A$.

11. If a matrix A rotates vectors counterclockwise by θ degrees, what does the matrix A^{-1} do?

The inverse matrix A^{-1} rotates vectors by θ degrees clockwise. In other words, it reverses the rotation of the matrix A .

12. If a 2×2 matrix has complex eigenvalues, what does this matrix do to vectors upon multiplication?

If a 2×2 matrix has complex eigenvalues $\lambda = a \pm bi$, it rotates vectors by an angle $\theta = \tan^{-1}(b/a)$ and stretches them by a factor $|\lambda| = \sqrt{a^2 + b^2}$.

13. Suppose A is a matrix and k is a positive integer. What does the notation A^k mean?

The notation A^k indicates that the matrix A is multiplied by itself k times, that is, $A^k = \underbrace{AA \cdots A}_{k \text{ times}}$.

- 14.** What is the relationship between the inverse of a matrix and the determinant of a matrix?

An $n \times n$ matrix A is invertible if and only if $\det A \neq 0$. That is, there exists a matrix inverse A^{-1} such that $AA^{-1} = I$, provided that $\det A \neq 0$.

- 15.** Explain what eigenvalues and eigenvectors are.

An eigenvalue is a scalar quantity λ that, together with a nonzero vector $\mathbf{v}$ called an eigenvector, satisfies the equation $A\mathbf{v} = \lambda\mathbf{v}$ where A is an $n \times n$ matrix. Consequently, an eigenvalue must satisfy the equation $\det(A - \lambda I) = 0$.

- 16.** Why does a 2×2 matrix have two eigenvalues?

The eigenvalues of a 2×2 matrix are determined by solving a quadratic characteristic polynomial. In general, there are two solutions to a quadratic equation, so there are two eigenvalues of a 2×2 matrix, though one of them may be repeated twice.

- 17.** Suppose a 2×2 matrix with real entries has complex eigenvalues. Why must the eigenvalues be complex conjugates?

In general, the eigenvalues of a 2×2 matrix satisfy a second-degree characteristic polynomial of the form $\lambda^2 + b\lambda + c = 0$ with solutions given by the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4c}}{2}$$

When the discriminant, $b^2 - 4c$, is negative, the solutions will

be complex and have the form $x = -\frac{b}{2} \pm \frac{\sqrt{4c - b^2}}{2}i$. Hence, these solutions are complex-conjugate pairs.

- 18.** Why is it sometimes useful to write a matrix A in the form $A = PDP^{-1}$, where D is a diagonal matrix?

The matrix decomposition $A = PDP^{-1}$ is useful when computing powers of the matrix A . For example, $A^k = PD^kP^{-1}$ and, since D is a diagonal matrix, D^k can be calculated by raising each entry of D to the power k . This provides a straightforward method for raising a matrix to a power.

- 19.** What does the Perron-Frobenius Theorem say, and why is it useful?

The Perron-Frobenius Theorem states that if A is a primitive $n \times n$ matrix whose entries are all nonnegative then

- There exists an eigenvalue of A , call it λ_1 , that is real and positive.
- λ_1 is greater in magnitude than all other eigenvalues.
- The components of the eigenvector associated with λ_1 are all positive.

The Perron-Frobenius Theorem can be used to determine the long-term behavior of the recursion $\mathbf{n}_{t+1} = A\mathbf{n}_t$ without explicitly finding a solution.

1. (a) What is a function of two variables?

It is a rule that assigns to each ordered pair of real numbers (x, y) in its domain a unique real number denoted by $f(x, y)$.

(b) Describe two methods for visualizing a function of two variables.

One way to visualize a function of two variables is by graphing it, resulting in the surface $z = f(x, y)$. Another method for visualizing a function of two variables is a contour map.

The contour map consists of level curves of the function, with equations $f(x, y) = k$, which are horizontal traces of the graph of the function projected onto the xy -plane.

2. What does

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$$

mean? How can you show that such a limit does not exist?

$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$ means the values of $f(x, y)$ approach the number L as the point (x, y) approaches the point (a, b) along any path that is within the domain of f . We can show that a limit at a point does not exist by finding two different paths approaching the point along which $f(x, y)$ has different limits.

3. (a) What does it mean to say that f is continuous at (a, b) ?

f is continuous at (a, b) if $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = f(a, b)$.

(b) If f is continuous on $\mathbb{R}^2$, what can you say about its graph?

If f is continuous on $\mathbb{R}^2$, its graph will appear as a surface without holes or breaks.

4. (a) Write expressions for the partial derivatives $f_x(a, b)$ and $f_y(a, b)$ as limits.

$$f_x(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h}$$

$$f_y(a, b) = \lim_{h \rightarrow 0} \frac{f(a, b + h) - f(a, b)}{h}$$

(b) How do you interpret $f_x(a, b)$ and $f_y(a, b)$ geometrically? How do you interpret them as rates of change?

The partial derivative $f_x(a, b)$ is the slope of the tangent line at the point $(a, b, f(a, b))$ to the vertical trace in the plane $y = b$, and $f_y(a, b)$ is the slope of the tangent line to the trace in the plane $x = a$. Also $f_x(a, b)$ represents the rate of change of f with respect to x (at $x = a$) when y is fixed ($y = b$), and $f_y(a, b)$ is the rate of change of f with respect to y (at $y = b$) when x is fixed ($x = a$).

(c) If $f(x, y)$ is given by a formula, how do you calculate f_x and f_y ?

To find f_x , regard y as a constant and differentiate $f(x, y)$ with respect to x . To find f_y , regard x as a constant and differentiate $f(x, y)$ with respect to y .

5. What does Clairaut's Theorem say?

Suppose f is defined on a disk D that contains the point (a, b) . If the functions f_{xy} and f_{yx} are both continuous on D , then $f_{xy}(a, b) = f_{yx}(a, b)$.

6. How do you find an equation for the tangent plane to a surface $z = f(x, y)$?

If f has continuous partial derivatives, then an equation of the tangent plane to the surface $z = f(x, y)$ at the point $P(x_0, y_0, z_0)$ is $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$.

7. Define the linearization of f at (a, b) . What is the corresponding linear approximation? What is the geometric interpretation of the linear approximation?

The linearization of f at (a, b) is

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

and the corresponding linear approximation is

$$f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

We can interpret the linearization of f at (a, b) geometrically as the linear function whose graph is the tangent plane to the graph of f at (a, b) . Thus it is the linear function that best approximates f near (a, b) .

8. (a) What does it mean to say that f is differentiable at (a, b) ?

The function $f(x, y)$ is differentiable at (a, b) if

$$\lim_{(x,y) \rightarrow (a,b)} \frac{|f(x, y) - L(x, y)|}{\sqrt{(x - a)^2 + (y - b)^2}} = 0$$

(b) How do you usually verify that f is differentiable?

We check that the partial derivatives f_x and f_y exist and are continuous in some region D . If this is the case, then f is differentiable in D .

9. State the Chain Rule for the case where $z = f(x, y)$ and x and y are functions of a variable t .

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

10. If z is defined implicitly as a function of x and y by an equation of the form $F(x, y, z) = 0$, how do you find $\partial z / \partial x$ and $\partial z / \partial y$?

Provided $\partial F / \partial z \neq 0$, the partial derivatives are given by

$$\frac{\partial z}{\partial x} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}} \quad \frac{\partial z}{\partial y} = -\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial z}}$$

11. (a) Write an expression as a limit for the directional derivative of f at (x_0, y_0) in the direction of a unit vector $\mathbf{u} = [a, b]$. How do you interpret it as a rate? How do you interpret it geometrically?

$$D_{\mathbf{u}}f(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}$$

We can interpret the directional derivative as the rate of change of f at (x_0, y_0) in the direction of $\mathbf{u}$. Geometrically, if P is the point $(x_0, y_0, f(x_0, y_0))$ on the graph of f and C is the curve of intersection of the graph of f with the vertical plane that passes through P in the direction $\mathbf{u}$, the directional derivative of f at (x_0, y_0) in the direction of $\mathbf{u}$ is the slope of the tangent line to C at P . (See Figure 9.5.3.)

- (b) If f is differentiable, write an expression for $D_{\mathbf{u}}f(x_0, y_0)$ in terms of f_x and f_y .

$$D_{\mathbf{u}}f(x, y) = f_x(x, y)a + f_y(x, y)b \quad \text{where } \mathbf{u} = [a, b]$$

12. (a) Define the gradient vector ∇f for a function f of two variables.

$$\nabla f(x, y) = [f_x(x, y), f_y(x, y)]$$

- (b) Express $D_{\mathbf{u}}f$ in terms of ∇f .

$$D_{\mathbf{u}}f(x, y) = \nabla f(x, y) \cdot \mathbf{u}$$

- (c) Explain the geometric significance of the gradient.

The gradient vector of a function points in the direction of maximum rate of increase of the function. On a graph of the function, the gradient points in the direction of steepest ascent.

13. What do the following statements mean?

- (a) f has a local maximum at (a, b) .

f has a local maximum at (a, b) if $f(x, y) \leq f(a, b)$ when (x, y) is near (a, b) .

- (b) f has an absolute maximum at (a, b) .

f has an absolute maximum at (a, b) if $f(x, y) \leq f(a, b)$ for all points (x, y) in the domain of f .

- (c) f has a local minimum at (a, b) .

f has a local minimum at (a, b) if $f(x, y) \geq f(a, b)$ when (x, y) is near (a, b) .

- (d) f has an absolute minimum at (a, b) .

f has an absolute minimum at (a, b) if $f(x, y) \geq f(a, b)$ for all points (x, y) in the domain of f .

- (e) f has a saddle point at (a, b) .

f has a saddle point at (a, b) if $f(a, b)$ is a local maximum in one direction but a local minimum in another.

14. (a) If f has a local maximum at (a, b) , what can you say about its partial derivatives at (a, b) ?

If f has a local maximum at (a, b) and the first-order partial derivatives of f exist there, then $f_x(a, b) = 0$ and $f_y(a, b) = 0$.

- (b) What is a critical point of f ?

A point (a, b) is a critical point of f if $f_x(a, b) = 0$ and $f_y(a, b) = 0$, or if one of these partial derivatives does not exist.

15. State the Second Derivatives Test.

Suppose that f_x and f_y are continuous on a disk with center (a, b) , and that $f_x(a, b) = 0$ and $f_y(a, b) = 0$. Let

$$D = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2$$

- If $D > 0$ and $f_{xx}(a, b) > 0$, then $f(a, b)$ is a local minimum.
- If $D > 0$ and $f_{xx}(a, b) < 0$, then $f(a, b)$ is a local maximum.
- If $D < 0$, then f has a saddle point at (a, b) .
- If $D = 0$, then the test gives no information.

16. (a) What is a closed set in $\mathbb{R}^2$? What is a bounded set?

A closed set in $\mathbb{R}^2$ is a set that contains all its boundary points. A bounded set in $\mathbb{R}^2$ is one that is contained within some disk. In other words, it is finite in extent.

- (b) State the Extreme Value Theorem for functions of two variables.

If f is continuous on a closed, bounded set D in $\mathbb{R}^2$, then f attains an absolute maximum value $f(x_1, y_1)$ and an absolute minimum value $f(x_2, y_2)$ at some points (x_1, y_1) and (x_2, y_2) in D .

- (c) How do you find the values that the Extreme Value Theorem guarantees?

To find the absolute maximum and minimum values of a continuous function f on a closed, bounded set D :

- Find the values of f at the critical points of f in D .
- Find the extreme values of f on the boundary of D .
- The largest of the values from steps 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

1. What is the difference between an autonomous and a non-autonomous system of differential equations?

A system of differential equations is autonomous if the equations do not explicitly contain the independent variable. If the differential equations depend on the independent variable, it is a non-autonomous system.

2. What is an equilibrium of a system of differential equations?

An equilibrium of the system of differential equations $d\mathbf{x}/dt = f(\mathbf{x})$ is a vector of values $\hat{\mathbf{x}}$ that satisfies the equation $d\mathbf{x}/dt = \mathbf{0}$.

3. Explain the difference between local and global stability in systems of differential equations.

An equilibrium $\hat{\mathbf{x}}$ of a system of differential equations is locally stable if $\mathbf{x}$ approaches the value $\hat{\mathbf{x}}$ as $t \rightarrow \infty$ for all initial values of $\mathbf{x}$ sufficiently close to $\hat{\mathbf{x}}$. The equilibrium is globally stable if the system approaches $\hat{\mathbf{x}}$ for all initial values of $\mathbf{x}$.

4. What is the difference between the solution of an initial-value problem and the general solution of a system of differential equations?

The general solution represents a *family* of solutions all of which satisfy the system of differential equations, whereas the solution of an initial-value problem is a *single* solution that satisfies the system of differential equations along with an initial condition $\mathbf{x}(t_0) = \mathbf{x}_0$.

5. What does the Existence and Uniqueness Theorem tell us about homogeneous systems of linear, autonomous differential equations?

The Existence and Uniqueness Theorem tell us that an initial-value problem involving a homogeneous systems of linear, autonomous differential equations will have one and only one solution $\mathbf{x}(t)$ and this solution is defined for all $t \in \mathbb{R}$.

6. Explain the Superposition Principle.

The Superposition Principle says that if $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ are solutions to the linear system of differential equations $d\mathbf{x}/dt = A\mathbf{x}$, then $\mathbf{x}(t) = c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t)$, where c_1 and c_2 are scalar quantities, is also a solution.

7. Explain the difference between nullclines and eigenvectors in systems of linear autonomous differential equations.

The nullclines of a system of linear differential equations are straight lines in the phase plane along which the rate of change of

one of the variables is zero. This means the solution curves move in either a vertical or horizontal direction along nullclines. Eigenvectors lie on straight lines that are themselves solution curves in the phase plane.

8. What is the linearization of a system of nonlinear differential equations?

The linearization of the system of nonlinear differential equations

$$\frac{dx_1}{dt} = f_1(x_1, x_2) \quad \text{and} \quad \frac{dx_2}{dt} = f_2(x_1, x_2)$$

is given by

$$\frac{d\boldsymbol{\epsilon}}{dt} = \begin{bmatrix} \frac{\partial f_1(\hat{x}_1, \hat{x}_2)}{\partial x_1} & \frac{\partial f_1(\hat{x}_1, \hat{x}_2)}{\partial x_2} \\ \frac{\partial f_2(\hat{x}_1, \hat{x}_2)}{\partial x_1} & \frac{\partial f_2(\hat{x}_1, \hat{x}_2)}{\partial x_2} \end{bmatrix} \boldsymbol{\epsilon}$$

where $\boldsymbol{\epsilon}$ is the vector whose components are the deviations of x_1 and x_2 from their equilibrium values.

9. Explain what a Jacobian matrix is.

The Jacobian matrix of the system of differential equations

$$\frac{dx_1}{dt} = f_1(x_1, x_2) \quad \text{and} \quad \frac{dx_2}{dt} = f_2(x_1, x_2)$$

is defined as

$$J(x_1, x_2) = \begin{bmatrix} \frac{\partial f_1(x_1, x_2)}{\partial x_1} & \frac{\partial f_1(x_1, x_2)}{\partial x_2} \\ \frac{\partial f_2(x_1, x_2)}{\partial x_1} & \frac{\partial f_2(x_1, x_2)}{\partial x_2} \end{bmatrix}$$

The Jacobian matrix can be used to determine the local stability properties of equilibria.

10. What do the eigenvalues of a Jacobian matrix from a system of nonlinear differential equations tell us?

The eigenvalues of a Jacobian matrix can be used to determine the local stability properties of an equilibrium $(\hat{x}_1, \hat{x}_2)$. If r is the largest eigenvalue of $J(\hat{x}_1, \hat{x}_2)$ when they are real, or the largest real part of the eigenvalues when they are complex, then the equilibrium is stable when $r < 0$ and unstable when $r > 0$.

